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Detecting Model Misspecification in Bayesian Inverse Problems via Variational Gradient Descent

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Abstract

Bayesian inference is optimal when the statistical model is well-specified, while outside this setting Bayesian inference can catastrophically fail; accordingly a wealth of post-Bayesian methodologies have been proposed. Predictively oriented (PrO) approaches lift the statistical model P_θ to an (infinite) mixture model $\int P_\theta \, \mathrm{d}Q(\theta)$ and fit this predictive distribution via minimising an entropy-regularised objective functional. In the well-specified setting one expects the mixing distribution Q to concentrate around the true data-generating parameter in the large data limit, while such singular concentration will typically not be observed if the model is misspecified. Our contribution is to demonstrate that one can empirically detect model misspecification by comparing the standard Bayesian posterior to the PrO 'posterior' Q, and that each can be efficiently computed via variational gradient descent. A simulation study confirms that model misspecification can be rigorously tested using this framework, and a detailed case study in seismic tomography illustrates the potential of this approach as a novel and widely-applicable diagnostic tool for the standard Bayesian workflow.

1 Introduction

Detecting and mitigating model misspecification is a pertinent but difficult issue in Bayesian statistics, where one must consider the full posterior predictive distribution *in lieu* of e.g. plotting a simple scalar residual. The two main approaches to detecting model misspecification (also called *model criticism*) are:

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1. *Predictive*: Compare the observed data to the posterior predictive distribution, either re-using the data that were used to construct it, as in posterior predictive checks [Rubin, 1984, Gelman et al., 1996, Bayarri and Berger, 2000], or using held-out data [Moran et al., 2024].

2. *Comparative*: Perform model selection over a set of candidates models in which the current model is contained [e.g. Kass and Raftery, 1995, Wasserman, 2000, Kamary et al., 2014].

In the first case, if the data are in some sense 'unexpected' under the posterior predictive distribution, this serves as evidence that either the prior or the model may be misspecified. When the same data are used both to construct the predictive distribution and to perform the check, the resulting posterior predictive p-values are known to be conservative; remedies include calibration by simulation [Hjort et al., 2006], modified p-values that remove the information in the test statistic from the posterior [Bayarri and Berger, 2000], and the use of held-out data [Moran et al., 2024]. In the second case, if the data strongly support an alternative model, this suggests that the original model may be misspecified. Of course, this is not a true dichotomy and the predictive and comparative approaches are intimately related; for instance, predictive performance is a common criteria for selecting a suitable model [Piironen and Vehtari, 2017, Fong and Holmes, 2020].

For challenging applications, both of these approaches can be impractical. An example of such a challenging application is *seismic travel time tomography* [Curtis and Snieder, 2002, Zhang and Curtis, 2020, Zhao et al., 2022], a regression task where one seeks to reconstruct a subsurface seismic velocity field using measured first arrival times of seismic waves travelling between seismic source and sensor (seismometer) locations. Evaluation of the likelihood and/or its gradient here is associated with a nontrivial computational cost, since the *Eikonal equation*, a partial differential equation describing the high frequency approximation of the wave equation, needs to be solved to estimate the travel times of the first arriving seismic waves. It is worth pointing out that 'misspecification' as discussed in this work refers to the suitability of the statistical model for the dataset, rather than any errors that may be present in the physical model for seismic wave travel, though the two issues are of course closely related.

Considering the predictive approach in the seismic tomography context, calibrating a check that re-uses the data requires the posterior to be recomputed for many synthetic datasets, each requiring repeated solution of the Eikonal equation, while for checks based on held-out data the spatiotemporal nature of the observations renders the data strongly dependent, representing a challenge in constructing a suitable held-out dataset. Further, while in principle there are solutions to cross-validation for dependent data [e.g. Burman et al., 1994, Rabinowicz and Rosset, 2022], the computational cost associated with performing multiple folds of held-out prediction can render this approach impractical.

Focussing instead on the comparative approach, constructing plausible alternative models requires modifying the physical assumptions underlying the seismic wave propagation. This in turn requires the implementation and optimisation of suitable numerical methods, which

demands considerable effort. In the case of model misspecification, there is particular interest in exploring *more sophisticated* models, but without advanced physical insight, constructing such alternatives might be impractical.

The aim of this paper is to propose a *simple, practical and general approach to detecting model misspecification in Bayesian statistics*, and we focus on the seismic travel time tomography problem to demonstrate its potential. Our solution can be considered both a predictive and a comparative approach; predictive because we explicitly assess the predictive performance of the model, and comparative because our principal innovation is to automate the generation of a candidate model set.

Inspired by *nonparametric maximum likelihood* [Laird, 1978], we consider lifting a statistical model P_θ to an (infinite) mixture model $P_Q := \int P_\theta \, \mathrm{d}Q(\theta)$ as a mechanism to generate an infinite candidate set of alternative models (parametrised by Q), while retaining any causal semantics present in the original statistical model. However, the nonparametric maximum likelihood estimate for Q is not computable in general. Instead, we therefore follow the *predictively oriented* (PrO) *posterior* approach of Lai et al. [2026], Shen et al. [2025], McLatchie et al. [2025], in which learning of the mixing density Q proceeds based on the performance of the predictive distribution P_Q (in this sense our method is a predictive approach) and is cast as an entropy-regularised optimisation task, with the solution being a PrO 'posterior' denoted Q_{PrO}. Our specific contributions are as follows:

- *Detecting misspecification:* We propose a direct comparison of the predictive distributions associated with Q_{PrO} and the Bayesian posterior Q_{Bayes} as a general strategy enabling model misspecification to be detected (in this sense our method is a comparative approach). Indeed, McLatchie et al. [2025] proved that in the well-specified setting the predictive distribution associated to Q_{PrO} converges to the true data-generating distribution in the large data limit, in agreement with the predictive distribution associated to Q_{Bayes}, while such agreement is typically not observed when the model is misspecified.

- *Testing for misspecification:* A formal hypothesis test for model misspecification is presented, and for regression models that are linear in the parameters we establish sufficient conditions for the resulting test to have asymptotically correct Type I error. Due to the parametric bootstrap, our test is practical only for statistical models P_θ for which both simulation and inference can be rapidly performed, motivating the development of efficient numerical methods to obtain Q_{PrO} and Q_{Bayes}.

- *Computation as variational gradient descent:* Since Q_{PrO} is defined as the minimiser of a *nonlinear* variational objective, we do not have access to an unnormalised form of the target, and as such standard methods such as Markov chain Monte Carlo (MCMC) cannot be immediately applied. As a solution, we turn to *variational gradient descent* [VGD; Wang and Liu, 2019, Chazal et al., 2025], which is a nonlinear generalisation of Stein variational gradient descent (SVGD) [Liu and Wang, 2016], a popular numerical method for computing Q_{Bayes}. Novel sufficient conditions for the consistency of VGD are presented, than can be verified in the seismology context. Remarkably, sampling

from Q_{PrO} can be achieved with a one-line change to standard SVGD, enabling both Q_{PrO} and Q_{Bayes} to be computed using identical code, with an additional argument specifying whether the PrO or Bayesian posterior is to be computed. Using VGD, we empirically confirm the ability of the parametric bootstrap hypothesis test to detect when the statistical model is misspecified.

- *Application to inverse problems:* For challenging settings where testing for misspecification using a parametric bootstrap would be computationally impractical, even using VGD, we investigate whether a visual comparison of Q_{PrO} and Q_{Bayes} can still act as a useful diagnostic tool. To this end a detailed case study involving seismic travel time tomography is presented. Here we indeed find that a visual comparison of Q_{PrO} and Q_{Bayes} is able to distinguish between the well-specified case and scenarios in which the location of the sensors is misspecified.

Our methods are contained in Section 2, while our empirical assessment is contained in Section 3. A summary of our findings is presented in Section 4, with our proofs and experimental protocol reserved for the Appendix.

2 Methods

The variational formulation of Bayesian updating and the related PrO approach are recalled in Section 2.1. The VGD methodology is presented in Section 2.2, and novel theoretical analysis required to establish its validity in our context is presented in Section 2.3. Our formal hypothesis test for misspecification is presented in Section 2.4 and sufficient conditions for the validity of the parametric bootstrap are presented in Section 2.5.

2.1 Bayesian and Predictively Oriented Approaches

Let P_θ denote a statistical model parametrised by $\theta \in \mathbb{R}^d$, whose density p_θ we assume to exist. Let $\mathfrak{D}_n$ denote the dataset. In what follows, motivated by our seismic tomography case study, we focus on regression modelling, where responses $\{y_i\}_{i=1}^n$ are conditionally independent given covariates $\{x_i\}_{i=1}^n$, so that

$$\log p_\theta(\mathfrak{D}_n) = \sum_{i=1}^n \log p_\theta(y_i|x_i)$$

where the dependence on the covariates x_i is made explicit. However it should be noted that our methods are applicable beyond the regression context.

Standard Bayesian Posterior Let $\mathcal{P}(\mathbb{R}^d)$ denote the set of distributions[1] on $\mathbb{R}^d$. Let $Q \ll Q_0$ denote that Q is absolutely continuous with respect to Q_0, and $\mathrm{d}Q/\mathrm{d}Q_0$ the

[1]Measurability is implicitly assumed in this manuscript.

Radon–Nikodym density of Q with respect to Q_0. For $Q \ll Q_0$, the Kullback–Leibler divergence (KLD) is defined as $\mathrm{KLD}(Q\|Q_0) := \int \log(\mathrm{d}Q/\mathrm{d}Q_0)\ \mathrm{d}Q$, while for $Q \not\ll Q_0$ we set $\mathrm{KLD}(Q||Q_0) = \infty$. Recall the variational characterisation of the standard Bayesian posterior due to Zellner [1988]:

$$Q_{\text{Bayes}} := \operatorname*{arg\,min}_{Q \in \mathcal{P}(\mathbb{R}^d)} -\sum_{i=1}^{n} \int \log p_\theta(y_i|x_i)\ \mathrm{d}Q(\theta) + \mathrm{KLD}(Q||Q_0) \tag{1}$$

where $Q_0 \in \mathcal{P}(\mathbb{R}^d)$ is the prior distribution [see also e.g. Knoblauch et al., 2022]. For the integral to be well-defined, i.e. for $\theta \mapsto -\log p_\theta(\mathfrak{D}_n)$ to be Q-integrable for all $Q \in \mathcal{P}(\mathbb{R}^d)$, it is sufficient for $\theta \mapsto p_\theta(\mathfrak{D}_n)$ to be bounded.

Predictively Oriented Posterior The PrO approaches of Lai et al. [2026], Shen et al. [2025], McLatchie et al. [2025] were developed with the aim of avoiding over-confident predictions when the statistical model is misspecified. These approaches lift the original parametric model P_θ into a mixture model P_Q, with density

$$p_Q(y_i|x_i) = \int p_\theta(y_i|x_i)\ \mathrm{d}Q(\theta),$$

and then attempt to learn Q by minimising an entropy-regularised objective functional. For the purpose of this paper we measure the suitability of Q using the (relative) entropy-regularised mixture log-likelihood

$$Q_{\text{PrO}} := \operatorname*{arg\,min}_{Q \in \mathcal{P}(\mathbb{R}^d)} -\sum_{i=1}^{n} \log p_Q(y_i|x_i) + \mathrm{KLD}(Q||Q_0). \tag{2}$$

This can be viewed as an entropy-regularised form of *nonparametric maximum likelihood* [Laird, 1978]; the entropic regularisation is a key ingredient, since otherwise the solution will be atomic [Lindsay, 1995, e.g. Theorem 21 in Chapter 5]. For a discussion of other related work, such as Masegosa [2020], Sheth and Khardon [2020], Jankowiak et al. [2020a,b], Morningstar et al. [2022], see Shen et al. [2025], McLatchie et al. [2025].

Since the variational formulation (2) is non-standard, we should first ask if Q_{PrO} is well-defined. Let $\mathcal{P}_\alpha(\mathbb{R}^d)$ denote the subset of $\mathcal{P}(\mathbb{R}^d)$ for which moments of order α exist. The proof of the following result is contained in Section A.2:

Theorem 1 (Q_{PrO} is well-defined)**.** *Let Q_0 admit a positive density q_0 on $\mathbb{R}^d$. Let $\theta \mapsto p_\theta(y_i|x_i)$ be positive, bounded and continuous for each (x_i, y_i) in the dataset. Then there exists a unique solution to* (2)*. Further, if $Q_0 \in \mathcal{P}_\alpha(\mathbb{R}^d)$ then $Q_{\text{PrO}} \in \mathcal{P}_\alpha(\mathbb{R}^d)$.*

The benefit of lifting to a mixture model is as follows: It the original statistical model P_θ was well-specified, so that there really was a correct parameter $\theta_\star$, then we can hope Q_{PrO} concentrates around $\theta_\star$ (i.e. collapses to a mixture model with a single mixture component). Likewise one would expect vanishing posterior uncertainty in the standard Bayesian context.

Inspecting whether the learned Q_{PrO} appears to converge to the same limit as the standard Bayesian posterior Q_{Bayes} can therefore provide a useful validation that the model is well-specified. On the other hand, if the original statistical model was misspecified, then we expect Q_{PrO} to learn a non-trivial mixture model, assuming such a mixture provides a better explanation of the dataset than any single instance of P_θ could. That is, Q_{PrO} is able to *adapt* to the level of model misspecification, in a way that standard Bayesian inference cannot. These intuitions for the asymptotic behaviour of Q_{PrO} are confirmed in the recent detailed theoretical treatment in McLatchie et al. [2025]. An empirical demonstration of the effectiveness of this approach is the subject of Section 3; the remainder of this section addresses the key practical question of how to calculate Q_{PrO} in (2).

Remark 1 (Comparison to mixture models)**.** *One can always ask whether a mixture model provides a better explanation of the data compared to any single instance of the original statistical model. The PrO posterior approach is fundamentally different to fitting a mixture model; there is no prior on the number of mixture components, and one does not need to extend the dimension of the parameter space as would ordinarily happen when mixture models are considered.*

Remark 2 (Choice of learning rate)**.** *Note that, unlike generalised Bayesian methods [Bissiri et al., 2016, Knoblauch et al., 2022] and in contrast to the earlier work on PrO approaches in Lai et al. [2026], Shen et al. [2025], McLatchie et al. [2025], no learning rate appears in* (2)*. Earlier works introduced a learning rate λ in the form of $\lambda \times \mathrm{KLD}(Q||Q_0)$ to accommodate other choices of data-dependent loss, such as maximum mean discrepancy. Since our aim is to assess the adequacy of the standard Bayesian analysis, $\lambda = 1$ is arguably a natural default, matching how KLD appears in the definition of the standard Bayesian posterior Q_{Bayes} in* (1)*, and for consistency we adopt the same value for Q_{PrO}. In fact, the methods that we develop in this work are remarkably insensitive to the choice of λ; see Remark 6. Selection of learning rates is known to be difficult [Wu and Martin, 2023] and it is therefore advantageous that this default does not need to be tuned.*

2.2 Variational Gradient Descent

An immediate question is *how to solve* (2)? Since the parameter of the mixture model is Q, it is unclear how to proceed; Q lives in $\mathcal{P}(\mathbb{R}^d)$ which is not a vector space, making it unclear how to apply operations such as taking a gradient with respect to Q. To resolve this problem we consider a general entropy-regularised variational objective

$$\mathcal{J}(Q) := \mathcal{L}(Q) + \mathrm{KLD}(Q||Q_0), \tag{3}$$

which accommodates both Q_{Bayes} and Q_{PrO} by taking the loss function to be, respectively, either

$$\mathcal{L}_{\mathrm{Bayes}}(Q) = -\sum_{i=1}^{n} \int \log p_\theta(y_i|x_i)\, \mathrm{d}Q(\theta), \tag{4}$$

for the standard Bayesian posterior, or

$$\mathcal{L}_{\mathrm{PrO}}(Q) = -\sum_{i=1}^{n} \log \int p_\theta(y_i|x_i)\, \mathrm{d}Q(\theta), \tag{5}$$

for the PrO posterior, which differ only in the order in which the integral and the logarithm are performed. Our aim is a rigorous notion of gradient descent that can be applied to (relative) entropy-regularised objective in (3). To this end we now explain how the VGD algorithm, due originally to Wang and Liu [2019], can be applied in our context.

Variational Gradient The notion of a gradient that we will need is a *variational gradient*. For a suitably regular functional $\mathcal{F} : \mathcal{P}(\mathbb{R}^d) \to \mathbb{R}$, the *first variation* at $Q \in \mathcal{P}(\mathbb{R}^d)$ is defined as a map $\mathcal{F}'(Q) : \mathbb{R}^d \to \mathbb{R}$ such that

$$\lim_{\epsilon \to 0} \frac{1}{\epsilon}\{\mathcal{F}(Q + \epsilon\chi) - \mathcal{F}(Q)\} = \int \mathcal{F}'(Q)\, \mathrm{d}\chi$$

for all perturbations χ of the form $\chi = Q' - Q$ with $Q' \in \mathcal{P}(\mathbb{R}^d)$; note that if it exists, the first variation is unique up to an additive constant. The first variation extends the standard derivative concept to arguments that are distribution-valued. Much of the intuition for standard derivatives carries over to the first variation; for example, a linear functional $\mathcal{F}_1(Q) = \int f(\theta)\, \mathrm{d}Q(\theta)$ has a first variation $\mathcal{F}_1'(Q)(\theta) = f(\theta)$ which is constant in Q. Similarly a quadratic functional $\mathcal{F}_2(Q) = \iint f(\theta, \vartheta)\, \mathrm{d}Q(\theta)\mathrm{d}Q(\vartheta)$, with f symmetric, has a first variation $\mathcal{F}_2'(Q)(\theta) = 2\int f(\theta, \vartheta)\, \mathrm{d}Q(\vartheta)$ which is linear in Q. (In both of these examples f must be regular enough for the first variation to be well-defined.)

Given a functional $\mathcal{F}(Q)$ we define the variational gradient of $\mathcal{F}$ at Q as the function $\nabla_{\mathrm{V}}\mathcal{F}(Q)(\theta) := \nabla_\theta \mathcal{F}'(Q)(\theta)$ where $\mathcal{F}'(Q)$ is the first variation of $\mathcal{F}$ at Q [Chazal et al., 2025, Definition 1]. The variational gradient generalises the better-known *Wasserstein gradient* from optimal transport beyond the Wasserstein space $Q \in \mathcal{P}_2(\mathbb{R}^d)$, so that we do not need to assume a second moment; see e.g. Otto [2001]. For the examples in the previous paragraph, again with f sufficiently regular, the variational gradients are $\nabla_{\mathrm{V}}\mathcal{F}_1(Q)(\theta) = \nabla_\theta f(\theta)$ and $\nabla_{\mathrm{V}}\mathcal{F}_2(Q)(\theta) = 2\int \nabla_\theta f(\theta, \vartheta)\, \mathrm{d}Q(\vartheta)$. For the loss functions in (4) and (5) we have

$$\nabla_{\mathrm{V}}\mathcal{L}_{\mathrm{Bayes}}(Q)(\theta) = -\sum_{i=1}^{n} \nabla_\theta \log p_\theta(y_i|x_i) \tag{6}$$

$$\nabla_{\mathrm{V}}\mathcal{L}_{\mathrm{PrO}}(Q)(\theta) = -\sum_{i=1}^{n} w_\theta^Q(y_i|x_i) \nabla_\theta \log p_\theta(y_i|x_i), \qquad w_\theta^Q(y_i|x_i) := \frac{p_\theta(y_i|x_i)}{p_Q(y_i|x_i)}, \tag{7}$$

see Proposition 2 in Section A.1. Note that (7) can be seen as a weighted version of (6), which agrees when $Q = \delta_\theta$ is a Dirac distribution at $\theta \in \mathbb{R}^d$.

Computing Directional Derivatives Let $T_{\#}Q$ denote the distribution of $T(X)$ where $X \sim Q$. Consider the directional derivatives

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon}\mathcal{J}((\mathrm{I}_d + \epsilon v)_{\#}Q)\bigg|_{\epsilon=0}$$

as specified by a suitable vector field $v : \mathbb{R}^d \to \mathbb{R}^d$, where I_d is the identity map on $\mathbb{R}^d$. For the purpose of optimisation, we seek a vector field v for which the rate of decrease in $\mathcal{J}$ is maximised. To this end, letting

$$\mathcal{T}_Q v(\theta) := [(\nabla \log q_0)(\theta) - \nabla_{\mathrm{V}}\mathcal{L}(Q)(\theta)] \cdot v(\theta) + (\nabla \cdot v)(\theta),$$

it follows from the fundamental theorem of calculus [see e.g. Section 3.2.2.2 of Chazal et al., 2025] that

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon}\mathcal{J}((\mathrm{I}_d + \epsilon v)_{\#}Q)\bigg|_{\epsilon=0} = -\int \mathcal{T}_Q v(\theta)\, \mathrm{d}Q(\theta). \tag{8}$$

That is, the directional derivative of the objective $\mathcal{J}$ in (3) can be expressed as an explicit Q-dependent linear functional applied to the vector field.

Following the Directions of Steepest Descent Next we pick the vector field v_Q from the unit ball of an appropriate Hilbert space for which the magnitude of the negative gradient in (8) is maximised. For a multivariate function, let $\partial_{i,j}$, ∇_i, etc, indicate the action of the differential operators with respect to the ith argument. Letting $\mathcal{H}_k$ denote the reproducing kernel Hilbert space (RKHS) associated to a symmetric positive semi-definite kernel $k : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$, we seek $v_Q \in \mathcal{H}_k^d$, the d-fold Cartesian product, which leads to

$$v_Q(\cdot) \propto \int \{k(\theta, \cdot)(\nabla \log q_0 - \nabla_{\mathrm{V}}\mathcal{L}(Q))(\theta) + \nabla_1 k(\theta, \cdot)\}\, \mathrm{d}Q(\theta). \tag{9}$$

To numerically approximate this gradient descent, we initialise $\{\theta_j^0\}_{j=1}^N$ as independent samples from μ_0 at time $t = 0$ and then update $\{\theta_j^t\}_{j=1}^N$ deterministically, via the coupled system of ordinary differential equations (ODEs)

$$\frac{\mathrm{d}\theta_i^t}{\mathrm{d}t} = \frac{1}{N}\sum_{j=1}^N k(\theta_i^t, \theta_j^t)(\nabla \log q_0 - \nabla_{\mathrm{V}}\mathcal{L}(Q_N^t))(\theta_j^t) + \nabla_1 k(\theta_j^t, \theta_i^t), \quad Q_N^t := \frac{1}{N}\sum_{j=1}^N \delta_{\theta_j^t} \tag{10}$$

up to a time horizon T. The first consistency result in this context was established in Proposition 3 of Chazal et al. [2025], who called the algorithm *variational gradient descent* (VGD). For Q_{Bayes}, VGD coincides with SVGD [Liu and Wang, 2016] and the sharpest convergence analysis of SVGD to-date appears in Banerjee et al. [2025]. Unfortunately the assumptions made in Chazal et al. [2025] are too restrictive to handle Q_{PrO}. To resolve this issue, a novel convergence guarantee for VGD in the context of Q_{PrO} is presented next.

Remark 3 (Other numerical methods for Q_{PrO})**.** *One could approximate Q_{PrO} using established numerical methods designed for variational tasks, the most well-studied of which is mean-field Langevin dynamics. However, the seismic tomography community have reported empirical evidence in favour of gradient flow-type algorithms such as SVGD over numerical methods based on long-run ergodic averages [see e.g. Zhang and Curtis, 2020, Zhang et al., 2023]. Here we propose to approximate both Q_{Bayes} and Q_{PrO} using VGD, noting that in the first case VGD coincides with SVGD due to the linear form of the loss function $\mathcal{L}_{\mathrm{Bayes}}$. In fact, we will see in Section 2.4 that applying VGD to Q_{PrO} requires only a one line-change to standard SVGD.*

2.3 Consistency of VGD

To discuss the consistency of VGD we need to specify in what sense the approximation is consistent. To this end, let $\mathcal{B}_k^d = \{v \in \mathcal{H}_k^d : \sum_i \|v_i\|_{\mathcal{H}_k}^2 \leq 1\}$ denote the unit ball in $\mathcal{H}_k^d$. Let $\mathcal{L}^1(Q) := \{f : \mathbb{R}^d \to \mathbb{R} : \int \|f(x)\| \, \mathrm{d}Q(x) < \infty\}$ denote[2] the set of Q-integrable functions on $\mathbb{R}^d$. Let f_- denote the negative part $f_- : x \mapsto \min\{0, f(x)\}$ of a function f. The kernel gradient discrepancy (KGD) [Chazal et al., 2025, Definition 4]

$$\mathrm{KGD}_k(Q) := \sup_{\substack{v \in \mathcal{B}_k^d \text{ s.t.} \\ (\mathcal{T}_Q v)_- \in \mathcal{L}^1(Q)}} \left| \int \mathcal{T}_Q v(\theta) \, \mathrm{d}Q(\theta) \right| \tag{11}$$

can be interpreted as a gradient norm for $\mathcal{J}$ using (8); a small KGD indicates that Q is close to being a stationary point (and in particular a minimiser, due to convexity) of $\mathcal{J}$. The precise topologies induced on $\mathcal{P}(\mathbb{R}^d)$ by KGD can be weaker or stronger depending on how the kernel k is selected; this is discussed in detail in Chazal et al. [2025] but such discussion is beyond the scope of the present work.

Let $C^r(\mathbb{R}^d)$ denote the set of r times continuously differentiable functionals on $\mathbb{R}^d$. The proof of the following result is contained in Section A.3:

Theorem 2 (Consistency of VGD for Q_{PrO})**.** *Assume that:*

(i) Initialisation: *μ_0 has bounded support, and has a density that is bounded.*

(ii) Kernel: *k is $C^2(\mathbb{R}^d \times \mathbb{R}^d)$ with the growth of $(\theta, \vartheta) \mapsto \|\nabla_1 k(\theta, \vartheta)\|$ at most linear, $\sup_\theta |\Delta_1 k(\theta, \theta)| < \infty$, and $\sup_\theta |\nabla_1 k(\theta, \theta) \cdot \nabla \log q_0(\theta)| < \infty$.*

(iii) Regularisation: *$\log q_0 \in C^2(\mathbb{R}^d)$ with the growth of $(\theta, \vartheta) \mapsto |k(\theta, \vartheta)| \|\nabla \log q_0(\vartheta)\|$ at most linear, and $\sup_\theta k(\theta, \theta) |\Delta \log q_0(\theta)| < \infty$.*

(iv) Regularity of P_θ: *$\theta \mapsto p_\theta(y_i|x_i)$ is positive, bounded and $C^2(\mathbb{R}^d)$ with*

a. $\sup_\theta \sqrt{k(\theta, \theta)} \dfrac{\|\nabla_\theta p_\theta(y_i|x_i)\|}{p_\theta(y_i|x_i)} < \infty$

[2]This should not be confused with the notation $\mathcal{L}$ for the loss function in (3).

Algorithm 1 Variational Gradient Descent for Q_{Bayes} and Q_{PrO}

Require: $\{\theta_j^0\}_{j=1}^N \subset \mathbb{R}^d$ (initial particles), $\epsilon > 0$ (step size)
  for $t = 0, \dots, T-1$ **do**
    $w_\theta^t(y_i|x_i) := p_\theta(y_i|x_i)/(\frac{1}{N}\sum_{r=1}^N p_{\theta_r^t}(y_i|x_i))$
    $s_t(\theta) := \begin{cases} (\nabla \log q_0)(\theta) + \sum_{i=1}^n \nabla_\theta \log p_\theta(y_i|x_i) & \text{to target } Q_{\mathrm{Bayes}} \\ (\nabla \log q_0)(\theta) + \sum_{i=1}^n w_\theta^t(y_i|x_i) \nabla_\theta \log p_\theta(y_i|x_i) & \text{to target } Q_{\mathrm{PrO}} \end{cases}$
    for $j = 1, \dots, N$ **do**
      $\theta_j^{t+1} \leftarrow \theta_j^t + \frac{\epsilon}{N}\sum_{r=1}^N \nabla_1 k(\theta_r^t, \theta_j^t) + s_t(\theta_r^t) k(\theta_r^t, \theta_j^t)$
    end for
  end for

b. $\sup_\theta k(\theta,\theta)\dfrac{|\Delta_\theta p_\theta(y_i|x_i)|}{p_\theta(y_i|x_i)} < \infty$

c. $\sup_\theta \|\nabla_1 k(\theta,\theta)\|\dfrac{\|\nabla_\theta p_\theta(y_i|x_i)\|}{p_\theta(y_i|x_i)} < \infty$

for each (x_i, y_i) in the dataset.

Then the dynamics defined in (10) *with $\mathcal{L} = \mathcal{L}_{\mathrm{PrO}}$ satisfies*

$$\frac{1}{T}\int_0^T \mathbb{E}[\mathrm{KGD}_k^2(Q_N^t)]\,\mathrm{d}t \le \frac{C_0}{T} + \frac{C_k}{N}$$

for some constants C_0 and C_k which are N and T-independent.

Theorem 2 provides the first consistency guarantee for VGD in the setting of $\mathcal{L}_{\mathrm{PrO}}$. The conditions in Theorem 2 involving $\nabla_1 k(\theta,\theta)$ are automatically satisfied when k is translation-invariant, such as the Gaussian and inverse multi-quadric kernels used in Section 3, since then $\theta \mapsto k(\theta,\theta)$ is constant and $\nabla_1 k(\theta,\theta) = 0$. For such bounded kernels the growth condition in (iii) reduces to at most linear growth of $\theta \mapsto \|\nabla \log q_0(\theta)\|$.

Remark 4 (On the assumptions)**.** *Our assumptions in Theorem 2 rule out models for which the* Fisher score $\theta \mapsto \nabla_\theta \log p_\theta(y_i|x_i)$ *is unbounded when the kernel is translation-invariant. This is a strong assumption in general, but it is likely to be satisfied in applications that are physics-constrained, since physical considerations often dictate that model parameters can be confined to a compact set (and then reparametrised back to $\mathbb{R}^d$), as done for seismic tomography in Zhang et al. [2023]. Establishing the consistency of VGD under weaker assumptions would be an interesting direction for further work.*

Remark 5 (Implementation of VGD)**.** *For the purposes of this paper, computation of both Q_{Bayes} and Q_{PrO} was performed using a time discretisation of VGD as described in Algorithm 1. In each case, Algorithm 1 starts by initialising N particles and then iteratively updating the particles according to an Euler discretisation of the ODEs* (10) *with step size*

$\epsilon > 0$ to be specified. After T time steps, the collection of particles represents an empirical approximation to either Q_{Bayes} or Q_{PrO}. It is worth reiterating that computation of Q_{PrO} requires a one-line change to existing implementations of SVGD, and that Q_{Bayes} and Q_{PrO} can be computed in parallel. The rapid convergence of VGD in a toy two-dimensional setting is displayed in Figure 1.

2.4 Testing for Misspecification

Our approach, in a nutshell, is to see if both Q_{Bayes} and Q_{PrO} appear to be converging to the same limit or not. If these distributions are substantially different, we interpret this as evidence that the model may be misspecified.

An obvious question at this point is *how to decide whether Q_{Bayes} and Q_{PrO} are converging to the same limit*? If the model is well-specified, Q_{Bayes} concentrates around the true parameter $\theta_\star$. Thus we are interested in whether Q_{PrO} also appears to concentrate around $\theta_\star$ or not. Unfortunately, it is not the case that Q_{Bayes} and Q_{PrO} concentrate at the same rate; it is well-known that Q_{Bayes} concentrates at a rate $n^{-1/2}$, while it appears that a slower rate is typical for Q_{PrO}. The lack of a complete understanding of the concentration of Q_{PrO} limits the extent to which the above question can be answered. Instead, we propose to compare the predictive distributions

$$P_{Q_{\text{Bayes}}}(\cdot|x) = \int P_\theta(\cdot|x)\, \mathrm{d}Q_{\text{Bayes}}(\theta) \qquad \text{and} \qquad P_{Q_{\text{PrO}}}(\cdot|x) = \int P_\theta(\cdot|x)\, \mathrm{d}Q_{\text{PrO}}(\theta),$$

which for simplicity we will denote in shorthand as P_{Bayes} and P_{PrO}, with the dependence on x left implicit. In the well-specified case, for a suitable discrepancy $\mathcal{D}_n$ (which we will see later can be weakly n-dependent), we should expect that $\mathcal{D}_n(P_{\text{PrO}}, P_{\text{Bayes}}) \to 0$ in an appropriate sense as $n \to \infty$ [see McLatchie et al., 2025, Theorem 1]. That is, if the number of data n is large enough then P_{PrO} and P_{Bayes} should be almost identical when the model is well-specified. Conversely, if Q_{PrO} does not concentrate around the same parameter $\theta_\star$ as Q_{Bayes}, then we can expect to detect this as an irreducible difference between the predictive distributions P_{PrO} and P_{Bayes}.

To operationalise this idea, suppose that $\{y_i\}_{i=1}^n \subset \mathbb{R}^p$ for some $p \in \mathbb{N}$ and let $\kappa : \mathbb{R}^p \times \mathbb{R}^p \to \mathbb{R}$ be a symmetric positive semi-definite kernel. Then we propose to construct an approximate null distribution for the (average squared) maximum mean discrepancy (MMD) test statistic associated with κ [Smola et al., 2007], i.e.

$$\mathcal{T}(\{(x_i, y_i)\}_{i=1}^n) \equiv \mathcal{D}_n(P_{\text{PrO}}, P_{\text{Bayes}}) := \frac{1}{n}\sum_{i=1}^n \text{MMD}^2_\kappa(P_{\text{PrO}}(\cdot|x_i), P_{\text{Bayes}}(\cdot|x_i)), \tag{12}$$

under the hypothesis that the statistical model is well-specified. To do so, we let θ_n be any strongly consistent estimator of $\theta_\star$ (for the experiments we report, we take θ_n to be the mean of Q_{Bayes}), and use a parametric bootstrap; i.e. repeatedly compute $\mathcal{T}(\{(x_i, \tilde{y}_i)\}_{i=1}^n)$ based on synthetic datasets where $\tilde{y}_i$ is simulated from $P_{\theta_n}(\cdot|x_i)$. The actual value of the test statistic (12) can then be compared to the bootstrap null distribution so-obtained. Theoretical support for this parametric bootstrap is provided in Section 2.5.

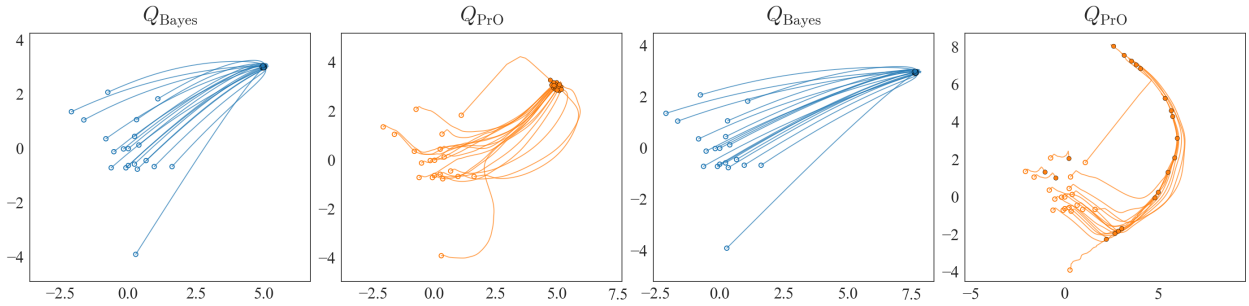


Figure 1: Illustrating the convergence of variational gradient descent (VGD) in the context of the two-dimensional example discussed in Section 3.1. The hollow circular markers depict initial particle locations $\{\theta_j^0\}_{j=1}^N$, lines represent trajectories at intermediate times $\{\theta_j^t\}_{j=1}^N$, and filled circular markers depict final locations $\{\theta_j^T\}_{j=1}^N$. The left and centre-left panels correspond to Q_{Bayes} and Q_{PrO} in a setting where the statistical model is well-specified, while the centre-right and right panels correspond to Q_{Bayes} and Q_{PrO} in a setting where the statistical model is misspecified. Here $N = 20$ particles were used.

Remark 6 (Insensitivity to the learning rate)**.** *Although no learning rate is required in* (2)*, it may be of interest to assess the sensitivity of the test to the relative weighting of the two terms in* (3)*. To this end, for* $\lambda > 0$ *one can replace* (3) *by* $\mathcal{J}_\lambda(Q) := \mathcal{L}(Q) + \lambda \,\mathrm{KLD}(Q||Q_0)$*, for both* $\mathcal{L}_{\text{Bayes}}$ *and* $\mathcal{L}_{\text{PrO}}$*. In particular, with* $\lambda = 1$*, the PrO objective differs from the Bayesian objective only through the interchange of the logarithm and expectation in the data-dependent loss, with the difference given by the non-negative Jensen gap. A sensitivity analysis can then be performed by computing the bootstrap p-value* p_λ *for different choices of* λ*. Our findings, reported in in Section B.5, show that the p-values* p_λ *are remarkably insensitive to even an order-of-magnitude variation away from our recommended default* $\lambda = 1$ *for each of the well-specified and misspecified settings that we considered.*

Remark 7 (Comparison to posterior predictive checks)**.** *A* posterior predictive check *compares the real dataset to synthetic datasets generated from* P_{Bayes} *[Rubin, 1984, Gelman et al., 1996]. Unlike our approach, no assumption that data are (conditionally) independent is needed to conduct a posterior predictive check. The approach can also be extended to a formal hypothesis test, using the parametric bootstrap to construct an empirical null in an analogous manner to that which we have described [Hjort et al., 2006]. However, upon failing a posterior predictive check, one may be left with limited guidance on how to build a more suitable model; at best we might hope to compare the posterior predictive performance of models from a given candidate set [Moran et al., 2023]. In contrast, if the statistical model is deemed to be misspecified, we immediately have the option of adopting* P_{PrO} *instead of* P_{Bayes} *as a viable predictive model (cf. Section 4).*

Remark 8 (Comparison to testing with mixtures)**.** *An approach to selecting among a collection of models* $\mathfrak{M}_i$*, proposed in Kamary et al. [2014], is to first fit a mixture* $\sum_i w_i \mathfrak{M}_i$ *and then consider the largest learned mixture weights* w_i *as the basis for selecting a best model. Though the authors focused on the setting where the true model is contained within*

the candidate model set, in the setting where all models are misspecified, it could occur that a non-trivial mixture is learned. This is similar in spirit to our approach if each $\mathfrak{M}_i$ represents an instance of the same original statistical model; however, fitting a mixture model is associated with substantial statistical and computational difficulties (cf. Remark 1), which our approach is able to avoid.

2.5 Consistency of the Parametric Bootstrap Null

This section presents sufficient conditions under which the parametric bootstrap, described in Section 2.4, yields a test whose Type I error is asymptotically correct. To state these conditions we need to be explicit about how the data are generated under the statistical model. To this end, let $P^{\theta,u}_{\text{Bayes}}$ and $P^{\theta,u}_{\text{PrO}}$ respectively denote the Bayesian and PrO predictive distributions based on the dataset arising from a parametrised *generator*

$$\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} G(\theta, x_1, u) \\ \vdots \\ G(\theta, x_n, u) \end{bmatrix}, \tag{13}$$

where $u \sim \nu$ is a random seed drawn from an appropriate reference distribution ν, and $x_1, \ldots, x_n$ are the covariates. Since the parametric bootstrap re-uses the observed covariates, throughout this section the covariates are held fixed, and all statements should be understood to hold for the given covariates $x_1, \ldots, x_n$. Let

$$\mathcal{T}_n(\theta, u) := \mathcal{D}_n(P^{\theta,u}_{\text{PrO}}, P^{\theta,u}_{\text{Bayes}})$$

denote the test statistic (12) computed from the dataset (13). If the statistical model is well-specified with true parameter $\theta_\star$, the observed value of the test statistic is $\mathcal{T}_n(\theta_\star, u)$ for some $u \sim \nu$, while the parametric bootstrap samples from the distribution of $\mathcal{T}_n(\theta_n, U)$, where $U \sim \nu$ is independent of the data and θ_n is held fixed at the value computed from the data. The test rejects the null hypothesis at level α when the observed value of the test statistic exceeds the $(1-\alpha)$-quantile $\hat{c}_n$ of this bootstrap distribution. The parametric bootstrap is therefore justified if the distribution of $\mathcal{T}_n(\theta, U)$ varies continuously with θ, in a manner that is uniform in n, and if θ_n is close enough to $\theta_\star$ relative to the scale on which $\mathcal{T}_n$ fluctuates.

To make this precise, let $\mathcal{H}_\kappa$ denote the RKHS of the kernel κ, and let $\mu_P := \int \kappa(y, \cdot)\, \mathrm{d}P(y) \in \mathcal{H}_\kappa$ denote the kernel mean embedding of a distribution P, so that $\mathrm{MMD}_\kappa(P, P') = \|\mu_P - \mu_{P'}\|_{\mathcal{H}_\kappa}$ [Smola et al., 2007]. Our analysis is restricted to regression models that are *linear in the parameters*, which include two of the three regression tasks considered in Section 3.1; the general case is discussed after Corollary 1. The proof of the following result is contained in Section A.5:

Theorem 3 (Stability of the Test Statistic for Linear Models)**.** *Assume that:*

(i) Linear model with additive noise: *The generator takes the form $G(\theta, x, u) = A(x)\theta + e(x, u)$, for some matrices $A(x) \in \mathbb{R}^{p \times d}$, where for each x the distribution of $e(x, u)$, $u \sim \nu$, has a positive, bounded and continuous density g_x, and correspondingly $p_\theta(y|x) = g_x(y - A(x)\theta)$.*

(ii) Prior with Lipschitz log-density: Q_0 *admits a positive density* q_0 *on* $\mathbb{R}^d$ *with* $|\log q_0(\theta) - \log q_0(\vartheta)| \leq L_0\|\theta - \vartheta\|$ *for some* $L_0 \geq 0$ *and all* θ, ϑ.

(iii) Bounded translation-invariant kernel: $\kappa(y + c, y' + c) = \kappa(y, y')$ *for all* y, y', $c \in \mathbb{R}^p$, *and* $M^2 := \sup_y \kappa(y, y) < \infty$.

Then, for all θ, ϑ, u, *all covariates* $x_1, \ldots, x_n$, *and all* n,

$$\left|\sqrt{\mathcal{T}_n(\vartheta, u)} - \sqrt{\mathcal{T}_n(\theta, u)}\right| \leq K\|\vartheta - \theta\|, \qquad K := 4ML_0.$$

In particular, the $(1-\alpha)$*-quantiles of* $\sqrt{\mathcal{T}_n(\vartheta, U)}$ *and* $\sqrt{\mathcal{T}_n(\theta, U)}$, $U \sim \nu$, *differ by at most* $K\|\vartheta - \theta\|$.

The constant K in Theorem 3 depends only on the prior and the kernel; it does not depend on n, on the covariates, on the random seed, on the matrices $A(x)$, or on the distribution of the noise. Thus the bootstrap null distribution is within $K\|\theta_n - \theta_\star\|$ of the true null distribution at the same sample size n, on the scale of $\sqrt{\mathcal{T}_n}$. This leads to the following sufficient condition for asymptotic Type I error control, whose proof is also contained in Section A.5:

Corollary 1 (Asymptotic Type I Error Control)**.** *Suppose that the statistical model is well-specified with true parameter* $\theta_\star$, *and that the assumptions of Theorem 3 hold. Let* $r_n > 0$ *be a sequence such that* $r_n\mathcal{T}_n(\theta_\star, U) \xrightarrow{d} F$, $U \sim \nu$, *where* F *is continuous and strictly increasing at its* $(1-\alpha)$*-quantile. Let* θ_n *be an estimator of* $\theta_\star$, *computed from the observed data, such that* $\sqrt{r_n}\|\theta_n - \theta_\star\| \xrightarrow{p} 0$. *Then the parametric bootstrap test has Type I error*

$$\mathbb{P}\left(\mathcal{T}_n(\theta_\star, u) > \hat{c}_n\right) \to \alpha$$

as $n \to \infty$, *where the randomness is with respect to the random seed* $u \sim \nu$ *that generated the observed data.*

For a root-n consistent estimator θ_n, such as the posterior mean of Q_{Bayes} under standard regularity conditions, the rate condition in Corollary 1 requires that $r_n = o(n)$; that is, $\sqrt{\mathcal{T}_n}$ must vanish more slowly than $n^{-1/2}$ under the null. Whether this holds depends on the rate at which Q_{PrO} concentrates relative to Q_{Bayes}, and on how this carries over to the predictive distributions, which is not yet well understood [McLatchie et al., 2025]. Establishing when the conditions of Corollary 1 hold is therefore left for future work.

The assumptions of Theorem 3 are satisfied by the Gaussian regression model (14) whenever the regression function is linear in the parameters, i.e. $f_\theta(x) = A(x)\theta$, together with the Gaussian kernel κ (for which $M = 1$) with a fixed lengthscale, and any prior whose log-density is Lipschitz, such as a multivariate Student-t or Laplace-type prior. This includes the quadratic and linear regression tasks in Section 3.1. Note that a Gaussian prior is not covered by assumption (ii), since its log-density is not Lipschitz. Extending the analysis beyond models that are linear in the parameters, such as the sigmoid regression task in Section 3.1 and the seismic tomography case study in Section 3.2, is left for future work.

3 Empirical Assessment

Preempting our application to seismic tomography, our focus in this section is on Gaussian regression models of the form

$$p_\theta(y_i|x_i) = \frac{1}{\sqrt{\det(2\pi\Sigma)}} \exp\left(-\frac{1}{2}\|\Sigma^{-\frac{1}{2}}(y_i - f_\theta(x_i))\|^2\right), \tag{14}$$

for responses $\{y_i\}_{i=1}^n$ conditional on covariates $\{x_i\}_{i=1}^n$, with known measurement error covariance matrix Σ and unknown regression parameters $\theta \in \mathbb{R}^d$. Our interest is in settings where the parametric regression function $f_\theta(x)$ could be misspecified. Proceeding with Bayesian inference would be problematic if $f_\theta(x)$ is indeed misspecified, since increasing the number of data n would cause the posterior to collapse onto a single 'best' parameter $\theta_\star$ and the predictions from the model will collapse also to $f_{\theta_\star}$, i.e. the predictions are simultaneously high-confidence and incorrect.

Our principal interest is in whether a comparison of Q_{Bayes} and Q_{PrO} based on a sufficiently large dataset enables misspecification to be detected. Section 3.1 reports a detailed empirical investigation in a controlled setting where data are generated from a simple known regression model, while a challenging application to seismic tomography is presented in Section 3.2. In both cases the statistical model takes the form (14), and we can interpret the sufficient conditions for the convergence of VGD in Theorem 2 this context. The proof of the following result is contained in Section A.4:

Proposition 1 (Regularity conditions for Gaussian regression model (14))**.** *Let P_θ be the Gaussian regression model in* (14)*, and let k be a symmetric positive semi-definite kernel for which $f_\theta(x_i)$, $\sqrt{k(\theta,\theta)}\nabla_\theta f_\theta(x_i)$, $\|\nabla_1 k(\theta,\theta)\|\|\nabla_\theta f_\theta(x_i)\|_{\text{op}}$, and $k(\theta,\theta)\Delta_\theta f_\theta(x_i)$ are bounded in θ for each x_i in the dataset. Then condition (iv) of Theorem 2 is satisfied.*

Although the assumptions of Proposition 1 are too strong for applications such as linear regression, where the regression function $f_\theta(x) = \theta \cdot x$ can be unbounded, for many physics-constrained inverse problems the boundedness requirements are likely to be satisfied.

3.1 Simulation Study

The aim of this section is to empirically assess whether our methods can detect when the regression function $f_\theta(x)$ is misspecified. To this end, we considered several toy regression tasks of the form (14), varying both the size of the dataset, the dimension of the parameter vector, and the extent to which the data-generating distribution departs from the regression model. Computation for these toy models is straightforward, so the formal test for model misspecification proposed in Section 2.4, based on the parametric bootstrap, is practical; this is in contrast to the seismic tomography case study in Section 3.2. Full details of our simulation setup are reserved for Section B.

Results are presented in Figure 2, where each row corresponds to a different regression model. It is visually apparent that the spread of the posterior predictive P_{PrO} is similar to

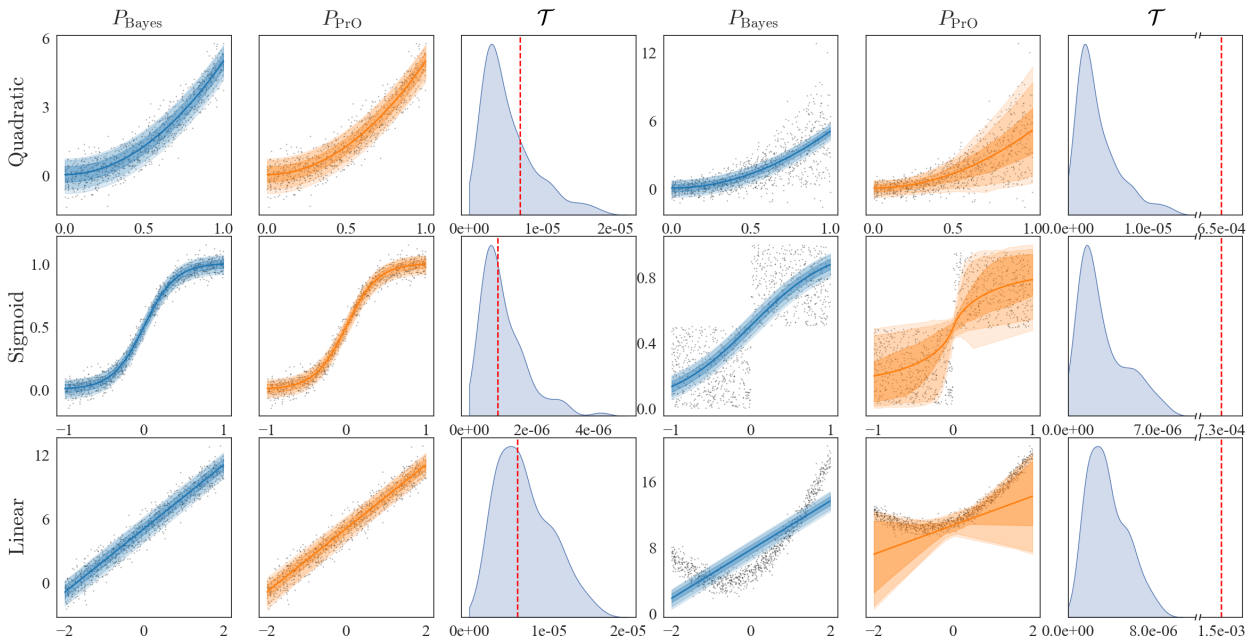


Figure 2: Simulation Study. Each row considers a regression task in which the data are either generated from the statistical model (well-specified, left) or not generated from the statistical model (misspecified, right). The posterior predictive distributions $P_{\text{Bayes}}(\cdot|x)$ (blue) and $P_{\text{PrO}}(\cdot|x)$ (orange) are displayed, along with the null distribution under the hypothesis that the statistical model is well-specified, and the actual realised value of the MMD test statistic $\mathcal{T}$ in (12) (red dashed).

that of P_{Bayes} when the model is well-specified, and much wider than the spread of P_{Bayes} when the model is misspecified. This difference in the misspecified setting occurs because the standard Bayesian posterior Q_{Bayes} is destined to concentrate on a single 'least bad' parameter $\theta_\star$ as the size of the dataset is increased, while the predictively oriented posterior Q_{PrO} is able to adapt to the level of model misspecification, resulting in an irreducible uncertainty in Q_{PrO} that does not vanish as the number of data is increased. The parameter posteriors Q_{Bayes} and Q_{PrO} themselves for these examples are presented in Figure 5 of Section B.4, where we also verify the convergence of the VGD algorithm used to obtain these results, as measured using the KGD in (11).

The distribution of the test statistic $\mathcal{T}$ under the parametric bootstrap null described in Section 2.4, alongside the actual realised value of $\mathcal{T}$, are also displayed in Figure 2. It can be seen that the realised value of $\mathcal{T}$ is far into the tail of the null distribution when data are not generated from the statistical model, meaning that the test statistic is able to detect that the statistical model is misspecified. Empirically, a larger sample size n increases the power of the test, as expected; see Figure 6 in Section B.4. Conversely, the detection of model misspecification is more challenging when a large number of parameters θ are being estimated; see Figure 7 in Section B.4.

3.2 Bayesian Seismic Travel Time Tomography

In seismic travel time tomography, the velocity structure of a medium (e.g., the Earth's subsurface) is estimated using measured first arrival times of seismic waves propagating between source and receiver locations [Curtis and Snieder, 2002, Zhang and Curtis, 2020, Zhao et al., 2022]. The parameter of interest is a scalar field, with $\theta(x)$ representing wave velocity[3] at a spatial location $x \in \mathbb{R}^3$. In practice, a bounded domain $\Omega \subset \mathbb{R}^3$ is discretised into a grid and the velocity field θ is represented as a vector of values associated to each cell in the grid, i.e. the velocity is modelled as being piecewise constant. The output of the regression model $f_\theta(x)$ represents the signal received by a seismometer at spatial location x, computed using a physics-constrained simulation using the velocity field θ. Typically, the model $f_\theta(x)$ is governed by the Eikonal equation $|\nabla_x f_\theta(x)| = \theta^{-1}(x)$, a high frequency approximation of the scalar wave equation, and is solved using the fast marching method [Rawlinson and Sambridge, 2005]. The measurement error covariance matrix Σ is assumed to be diagonal Zhao et al. [2022], with diagonal entries σ_i^2, where σ_i is set equal to 2% of the data associated to the ith channel.

In contrast to the examples in Section 3.1, the need to solve a partial differential equation here to evaluate the statistical model poses a computational barrier to investigating model misspecification using a hypothesis test. As such, here we content ourselves with a qualitative exploration of whether a visual comparison of Q_{Bayes} and Q_{PrO} can serve as a useful diagnostic for model misspecification in this challenging context. Of course, it is known that the regression model f_θ is to some extent misspecified relative to real world physics. For example, it represents a high frequency approximation of the wave physics, whereas the real case is band-limited. Additionally, travel time tomography relies only on kinematic (phase) information and ignores dynamic (amplitude) information, limiting its ability to reconstruct high resolution velocity structures of the Earth's interior. However, the key question here is whether the *statistical* model is misspecified in a way that could be scientifically consequential. To assess this, we consider a synthetic test-bed so that we have precise control of exactly how the data are misspecified.

For simplicity, our test-bed is defined for $x \in \mathbb{R}^2$ and is illustrated in the left panel of Figure 3. Here the seismic velocity field θ is piecewise constant, with $\theta(x) = 1$ km/s for $\|z\| \leq 2$, and $\theta(x) = 2$ km/s for $\|z\| > 2$. Data are obtained by first emitting a seismic wave from one of the 16 sensors (depicted by triangles in Figure 3) and recording the time at which the wave is detected at each of the other sensors, as calculated using the fast marching method. Iterating over all 16 sensors yields $n = 256$ readings, each measured with noise governed by the covariance matrix Σ. Since the error model is fixed and known, the measurement error component of the statistical model is always well-specified (see the centre panel of Figure 3). To simulate a realistic source of model misspecification we consider a setting where the data were generated using the correct sensor placement (white triangles in the right panel of Figure 3) while the statistical model assumes an incorrect sensor placement (red triangles). This form of model misspecification is common in earthquake

[3]In geophysics it is traditional to refer to speed, i.e. the magnitude of the velocity, simply as *velocity*.

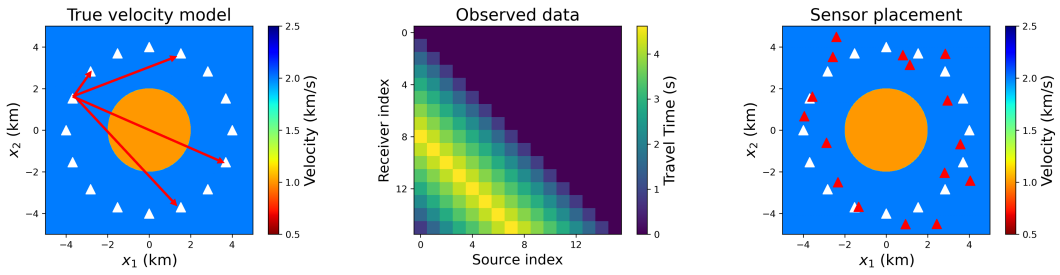
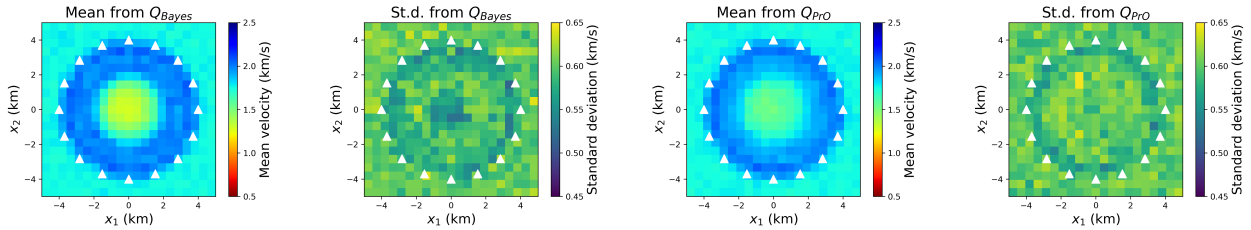


Figure 3: Seismic travel time tomography test-bed. Left: Data are obtained by first emitting a seismic wave from one of the 16 sensors (white triangles) and recording the time at which the wave is detected at each of the other sensors (left panel). Iterating over all sensors results in $n = 256$ travel times, which are noisily observed (centre panel). To simulate a realistic source of model misspecification we consider a setting where the data were generated using the correct sensor placement (white triangles in the right panel) while the statistical model assumes an incorrect sensor placement (red triangles).

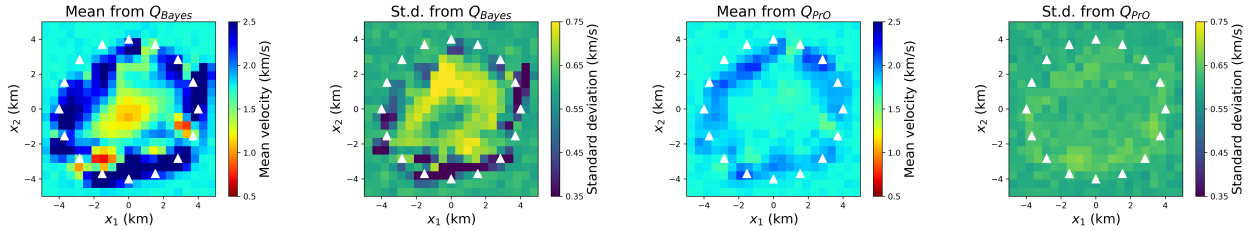
seismological tomography problems, in which the estimation of earthquake source locations can be inaccurate [Dziewonski et al., 1981], affecting the subsequent seismic tomography results.

To facilitate tomographic reconstruction, the spatial domain $\Omega = [-5, 5]^2$ is discretised into a 21×21 grid (the velocity field θ has dimension $d = 441$). The prior distribution Q_0 has each grid cell independent and uniformly distributed over the interval $[0.5, 3]$ km/s; in practice we reparametrised θ from $[0.5, 3]^{21\times 21}$ to $\mathbb{R}^{21\times 21}$ to avoid boundary considerations in VGD. Note that the data are non-informative about the region outside the convex hull of the sensors; accordingly the focus is on the reconstruction within the interior of the convex hull. Full details are contained in Section B.6.

Results are shown in Figure 4, where we compare pointwise means and standard deviations for Q_{Bayes} and Q_{PrO}. In the well-specified setting, the distributions Q_{Bayes} and Q_{PrO} appear almost identical; although Q_{PrO} may in theory concentrate more slowly than Q_{Bayes}, the difference between these two 'posteriors' cannot be easily distinguished, and certainly there is no evidence to suggest that these 'posteriors' are not converging to the same limit. In contrast, under model misspecification, clear differences between Q_{Bayes} and Q_{PrO} can be observed, both in the mean and standard deviation of the reconstructed velocity field. Specifically, Q_{Bayes} is over-confident regarding reconstruction near to the sensors, while such over-confidence is mitigated in Q_{PrO}. (Curiously, Q_{Bayes} also exhibits a *higher* standard deviation than Q_{PrO} for the central region; this indicates to us that reasoning about the impact of model misspecification on Q_{Bayes} is nontrivial.) These results are consistent with the interpretation that comparing Q_{Bayes} and Q_{PrO} can be an effective tool for detecting when a statistical model is misspecified. Indeed, the computational cost of producing the diagnostic plot in Figure 4 is only a factor of 2 larger than the cost of computing Q_{Bayes} itself.



(a) Well-specified sensor placement



(b) Misspecified sensor placement

Figure 4: Estimated seismic velocity θ in the setting where the sensor placement assumed in the statistical model is (a) well-specified and (b) misspecified. The standard Bayesian posterior Q_{Bayes} (left) and the predictively oriented posterior Q_{PrO} (right) are almost identical when the statistical model is well-specified, but differ substantially when the statistical model is misspecified.

4 Discussion

As statisticians we seek to avoid making predictions which are simultaneously highly confident and incorrect, but this scenario occurs generically in Bayesian analyses when the data are informative and the statistical model is misspecified. To address this challenge, we combined the emerging ideas of predictively oriented inference and variational gradient descent to obtain a simple, practical and general approach to detect model misspecification in the Bayesian context. An appealing aspect of our approach is that we do not require data splitting (e.g. as in holdout predictive checks or cross-validation) or the manual specification of alternative statistical models (as in comparative approaches). The approach was successfully demonstrated both in simulation studies and in the challenging seismic travel time tomography context.

In settings where our methods detect model misspecification, an obvious next question is *how to proceed with a misspecified model*? This important question is outside the scope of the present work, but has been the subject of considerable research effort. Potential solutions include nonparametric learning of the correct model [Kennedy and O'Hagan, 2001, Alvarez et al., 2013] and judicious use of an incorrect model [Bissiri et al., 2016, Knoblauch et al., 2022]; a recent review is provided by Nott et al. [2023]. However, a compelling alternative that is perfectly aligned with our work is to use the predictively oriented 'posterior' in place of the standard Bayesian posterior, as argued in Lai et al. [2026], Shen et al. [2025], McLatchie et al. [2025].

An important related issue is how to pinpoint which part(s) of the model are misspecified, in settings where misspecification is detected. Indeed, this would help to facilitate iterative model improvement, in-line with the standard Bayesian workflow [Gelman et al., 2020, Section 7]. The concept of a "part" only makes sense if additional structure is assumed on the statistical model; understanding how "local" misspecification gives rise to an irreducible uncertainty in the corresponding relevant parameters in Q_{PrO} would be an interesting direction for future work. Finally, our theoretical analysis of the parametric bootstrap was restricted to regression models that are linear in the parameters; extending this analysis to general models represents an important direction for future work.

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A Proofs

This appendix contains proofs for all theoretical results in the main text. Preliminary results are contained in Section A.1, while Theorem 1 is proven in Section A.2, Theorem 2 is proven in Section A.3, Proposition 1 is proven in Section A.4, and Theorem 3 and Corollary 1 are proven in Section A.5.

Additional Notation Let $b_Q(\theta) := (\nabla \log q_0)(\theta) - \nabla_{\mathrm{V}}\mathcal{L}(Q)(\theta)$ for all $Q \in \mathcal{P}(\mathbb{R}^d)$ and all $\theta \in \mathbb{R}^d$. This can be considered a Q-dependent generalisation of the *Stein score* [Liu and Wang, 2016], which is recovered in the case of linear $\mathcal{L}$. For $f, g : \mathbb{R}^d \to \mathbb{R}$, write $f(x) \lesssim g(x)$ if there exists a finite constant C such that $f(x) \leq Cg(x)$ for all $x \in \mathbb{R}^d$. For a suitably differentiable function f, let $\nabla^2 f$ denote the matrix of mixed partial derivatives $\partial_i \partial_j f$.

Throughout, $\|Q - Q'\|_{\mathrm{TV}}$ denotes the total variation norm of the signed measure $Q - Q'$, for which Pinsker's inequality states that $\|Q - Q'\|_{\mathrm{TV}} \leq \sqrt{2\mathrm{KLD}(Q||Q')}$.

A.1 Preliminary Results

First we derive the variational gradient of $\mathcal{L}_{\mathrm{PrO}}$:

Proposition 2 (Explicit form of variational gradient)**.** *Let $\mathcal{L} = \mathcal{L}_{\mathrm{PrO}}$. Let $\theta \mapsto p_\theta(y_i|x_i)$ be positive, bounded and differentiable for each (x_i, y_i) in the dataset. Then*

$$\nabla_{\mathrm{V}}\mathcal{L}(Q)(\theta) = -\sum_{i=1}^{n} \frac{\nabla_\theta p_\theta(y_i|x_i)}{p_Q(y_i|x_i)}. \tag{15}$$

Proof. The first variation is

$$\mathcal{L}'(Q)(\theta) = -\sum_{i=1}^{n} \frac{p_\theta(y_i|x_i)}{p_Q(y_i|x_i)} \tag{16}$$

which is well-defined since, under our assumptions, $p_Q(y_i|x_i) = \int p_\theta(y_i|x_i)\, \mathrm{d}Q(\theta)$ is strictly positive (so the denominator in (16) is non-zero) and bounded (so (16) is integrable with respect to all perturbations $\chi \in \mathcal{P}(\mathbb{R}^d)$; cf. the definition of first variation in Section 2.2) for all $Q \in \mathcal{P}(\mathbb{R}^d)$ and all (x_i, y_i) in the dataset. The variational gradient is thus (15), where the derivatives were assumed to be well-defined. □

The following specialises Proposition 1 in Chazal et al., 2025, which deals with general matrix-values kernels $K(\theta, \vartheta)$ to the case of a scalar kernel, i.e. $K(\theta, \vartheta) = k(\theta, \vartheta) I_{d\times d}$, and presents an explicit formula for the KGD.

Proposition 3 (Computable form of KGD; special case of Proposition 1 in Chazal et al., 2025)**.** *Let Q_0 have a density $q_0 > 0$ on $\mathbb{R}^d$. Let b_Q be well-defined. Let k be a symmetric*

positive semi-definite kernel for which $\nabla_1 k$ *and* $\nabla_1 \cdot \nabla_2 k$ *are well-defined. Suppose* $\mathcal{T}_Q \mathcal{H}_k^d \subset \mathcal{L}^1(Q)$*. Then*

$$\mathrm{KGD}_k(Q) = \left(\iint k_Q(\theta, \vartheta) \, \mathrm{d}Q(\theta) \mathrm{d}Q(\vartheta) \right)^{1/2}$$

where

$$k_Q(\theta, \vartheta) := \nabla_1 \cdot \nabla_2 k(\theta, \vartheta) + \nabla_1 k(\theta, \vartheta) \cdot b_Q(\vartheta) + \nabla_2 k(\theta, \vartheta) \cdot b_Q(\theta) + k(\theta, \vartheta) b_Q(\theta) \cdot b_Q(\vartheta)$$

is a Q*-dependent kernel.*

In particular, for $Q_N = \frac{1}{N} \sum_{i=1}^N \delta_{\theta_i}$ an empirical distribution,

$$\mathrm{KGD}_k^2(Q_N) = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N \left\{ \begin{array}{l} \nabla_1 \cdot \nabla_2 k(\theta_i, \theta_j) + \nabla_1 k(\theta_i, \theta_j) \cdot b_{Q_N}(\theta_j) \\ \quad + \nabla_2 k(\theta_i, \theta_j) \cdot b_{Q_N}(\theta_i) + k(\theta_i, \theta_j) b_{Q_N}(\theta_i) \cdot b_{Q_N}(\theta_j) \end{array} \right\} \quad (17)$$

which allows for KGD to be explicitly computed.

A.2 Proof of Theorem 1

Proof of Theorem 1. Introduce the shorthand

$$\mathcal{J}(Q) = \mathcal{L}(Q) + \mathrm{KLD}(Q||Q_0), \qquad \mathcal{L}(Q) = \mathcal{L}_{\mathrm{PrO}}(Q) = -\sum_{i=1}^n \log p_Q(y_i|x_i).$$

Under our assumptions $\mathcal{L}$ is weakly continuous, since if $Q_n \to Q$ weakly then

$$\int p_\theta(y_i|x_i) \, \mathrm{d}Q_n(\theta) \to \int p_\theta(y_i|x_i) \, \mathrm{d}Q(\theta)$$

since the integrand is bounded and continuous, while the limit is strictly positive since the integrand is positive. Thus

$$\mathcal{L}(Q_n) = -\sum_{i=1}^n \log \int p_\theta(y_i|x_i) \, \mathrm{d}Q_n(\theta) \to -\sum_{i=1}^n \log \int p_\theta(y_i|x_i) \, \mathrm{d}Q(\theta) = \mathcal{L}(Q),$$

as claimed. Further note that $\mathcal{L}$ is convex, since for $Q, Q' \in \mathcal{P}(\mathbb{R}^d)$ and $t \in (0, 1)$,

$$\begin{aligned} \mathcal{L}(tQ + (1-t)Q') &= -\sum_{i=1}^n \log \left[t \int p_\theta(y_i|x_i) \, \mathrm{d}Q(\theta) + (1-t) \int p_\theta(y_i|x_i) \, \mathrm{d}Q'(\theta) \right] \\ &\leq -t \sum_{i=1}^n \log \int p_\theta(y_i|x_i) \, \mathrm{d}Q(\theta) - (1-t) \sum_{i=1}^n \log \int p_\theta(y_i|x_i) \, \mathrm{d}Q'(\theta) \\ &= t\mathcal{L}(Q) + (1-t)\mathcal{L}(Q') \end{aligned}$$

by convexity of $z \mapsto -\log(z)$. Since $\mathrm{KLD}(\cdot||Q_0)$ is weakly lower semi-continuous and strictly convex, it follows that $\mathcal{J}$ is as well. The rest of the proof follows a standard argument [e.g. Proposition 5 in Hu et al., 2021]. To this end, note that

$$\mathcal{S} := \left\{ Q \in \mathcal{P}(\mathbb{R}^d) : \mathrm{KLD}(Q||Q_0) \leq \mathcal{J}(Q_0) - \inf_{Q' \in \mathcal{P}(\mathbb{R}^d)} \mathcal{L}(Q') \right\}.$$

Here the threshold is finite, since $\mathcal{J}(Q_0) = \mathcal{L}(Q_0) < \infty$ (as $p_{Q_0}(y_i|x_i) > 0$) and $\inf_{Q'} \mathcal{L}(Q') \geq -\sum_{i=1}^n \log \sup_\theta p_\theta(y_i|x_i) > -\infty$ (as $p_Q(y_i|x_i) \leq \sup_\theta p_\theta(y_i|x_i)$ for all Q). The set $\mathcal{S}$ is a (non-empty, since $Q_0 \in \mathcal{S}$) sub-level set of the KLD and is therefore weakly compact [Dupuis and Ellis, 2011, Lemma 1.4.3]. Since $\mathcal{J}$ is weakly lower semi-continuous, the minimum of $\mathcal{J}$ on $\mathcal{S}$ is attained. Since $\mathcal{J}(Q) \geq \mathcal{J}(Q_0)$ for all $Q \notin \mathcal{S}$, the minimum of $\mathcal{J}$ on $\mathcal{S}$ coincides with the global minimum of $\mathcal{J}$. Since $\mathcal{J}$ is strictly convex, the minimum is unique. The final claim is the content of Proposition 4. □

Proposition 4 (Moments for Q_{PrO})**.** *Assume that $Q_0 \in \mathcal{P}_\alpha(\mathbb{R}^d)$ admits a positive density q_0 on $\mathbb{R}^d$. Let $\theta \mapsto p_\theta(y_i|x_i)$ be positive, bounded and continuous for each (x_i, y_i) in the dataset. Then $Q_{\mathrm{PrO}} \in \mathcal{P}_\alpha(\mathbb{R}^d)$.*

Proof. Following the same argument used to prove Proposition 2, the first variation $\mathcal{L}'_{\mathrm{PrO}}$ is well-defined. Since Q_{PrO} is the minimiser of $\mathcal{J}$, following Chazal et al. [2025, Corollary 2] it is also a solution of the stationary point equation

$$\mathrm{cst} = \mathcal{L}'_{\mathrm{PrO}}(Q) + 1 + \log \frac{\mathrm{d}Q}{\mathrm{d}Q_0}$$

which implies Q_{PrO} has a density $q_{\mathrm{PrO}}(\theta) \propto \exp(-\mathcal{L}'_{\mathrm{PrO}}(Q_{\mathrm{PrO}})(\theta))q_0(\theta)$ on $\mathbb{R}^d$. From (16) and our assumption, $\mathcal{L}'_{\mathrm{PrO}}(Q_{\mathrm{PrO}})$ is bounded. The conclusion therefore follows from the assumption $Q_0 \in \mathcal{P}_\alpha(\mathbb{R}^d)$. □

A.3 Proof of Theorem 2

The main idea behind the proof of Theorem 2 is to undertake a refinement of the analysis of VGD in Chazal et al. [2025]; we do this in Section A.3.1. Then we verify that our refined regularity conditions hold in Section A.3.2, enabling the proof of Theorem 2 to be presented in Section A.3.3.

A.3.1 Refined Analysis of VGD

The following result relaxes the conditions of Proposition 3 in Chazal et al. [2025], to obtain a result that is applicable in our context; further details can be found in Remark 9. Note that Proposition 3 in Chazal et al. [2025] is in turn a generalisation (to VGD) of the analysis of SVGD presented in Theorem 1 of Banerjee et al. [2025].

For a sufficiently regular $\mathcal{F} : \mathcal{P}(\mathbb{R}^d) \to \mathbb{R}$, let $\nabla_{\mathrm{V}} \cdot \nabla_{\mathrm{V}} \mathcal{F}(Q)$ denote the function $(x, y) \mapsto \sum_i \partial_i \mathcal{G}'_{x,i}(Q)(y)$ where $\mathcal{G}_x(Q) := \nabla_{\mathrm{V}} \mathcal{F}(Q)(x)$.

Proposition 5 (Refined analysis of VGD)**.** *Assume that:*

(i) Lower bound and chain rule: $\mathcal{L}_{\min} := \inf_{Q\in\mathcal{P}(\mathbb{R}^d)} \mathcal{L}(Q) > -\infty$, $\mathbb{E}_{\theta_1,\dots,\theta_N\sim\mu_0}[\mathcal{L}(Q_N)] < \infty$, *and the map* $(\theta_1,\dots,\theta_N) \mapsto \mathcal{L}(Q_N)$ *is* $C^1(\mathbb{R}^{d\times N})$ *with* $\nabla_{\theta_i}\mathcal{L}(Q_N) = \frac{1}{N}\nabla_{\mathrm{V}}\mathcal{L}(Q_N)(\theta_i)$ *for each* $i \in \{1,\dots,N\}$,

(ii) Loss: *for each* $Q \in \mathcal{P}(\mathbb{R}^d)$, *the first variation* $\mathcal{L}'(Q)$ *exists and is* $C^2(\mathbb{R}^d)$, *and* $\nabla_{\mathrm{V}}\cdot\nabla_{\mathrm{V}}\mathcal{L}(Q)$ *exists; the map* $(\boldsymbol{\theta},\vartheta) \mapsto \nabla_{\mathrm{V}}\mathcal{L}(Q_N)(\vartheta)$ *is* $C^1(\mathbb{R}^{d\times N}\times\mathbb{R}^d)$, *where* $\boldsymbol{\theta} = (\theta_1,\dots,\theta_N)$ *and* $Q_N \equiv Q_N(\boldsymbol{\theta})$, *with* $\nabla_{\theta_i}\cdot[\nabla_{\mathrm{V}}\mathcal{L}(Q_N)(\vartheta)] = \frac{1}{N}\nabla_{\mathrm{V}}\cdot\nabla_{\mathrm{V}}\mathcal{L}(Q_N)(\vartheta)(\theta_i)$ *for each* $i \in \{1,\dots,N\}$ *and* $\vartheta \in \mathbb{R}^d$; *and*

  a. $\sup_{Q\in\mathcal{P}(\mathbb{R}^d)} |\int k(\theta,\theta)\nabla\cdot\nabla_{\mathrm{V}}\mathcal{L}(Q)(\theta)\,\mathrm{d}Q(\theta)| < \infty$,

  b. $\sup_{Q\in\mathcal{P}(\mathbb{R}^d)} |\int \nabla_1 k(\theta,\theta)\cdot\nabla_{\mathrm{V}}\mathcal{L}(Q)(\theta)\,\mathrm{d}Q(\theta)| < \infty$,

  c. $\sup_{Q\in\mathcal{P}(\mathbb{R}^d)} |\iint k(\theta,\vartheta)\nabla_{\mathrm{V}}\cdot\nabla_{\mathrm{V}}\mathcal{L}(Q)(\theta)(\vartheta)\,\mathrm{d}Q(\theta)\mathrm{d}Q(\vartheta)| < \infty$

(iii) Regularisation: $\log q_0 \in C^2(\mathbb{R}^d)$ *with* $\sup_\theta k(\theta,\theta)|\Delta\log q_0(\theta)| < \infty$,

(iv) Initialisation: μ_0 *has bounded support, and has a density that is bounded.*

(v) Kernel: k *is* $C^2(\mathbb{R}^d\times\mathbb{R}^d)$ *and satisfies both* $\sup_\theta |\Delta_1 k(\theta,\theta)| < \infty$ *and* $\sup_\theta |\nabla_1 k(\theta,\theta)\cdot\nabla\log q_0(\theta)| < \infty$.

(vi) Growth: *the maps* $(\theta_1,\dots,\theta_N) \mapsto |k(\theta_i,\theta_j)|\|\nabla\log q_0(\theta_j)\|$, $|k(\theta_i,\theta_j)|\|\nabla_{\mathrm{V}}\mathcal{L}(Q_N)(\theta_j)\|$ *and* $\|\nabla_1 k(\theta_j,\theta_i)\|$ *have at most linear growth for each* $\{i,j\} \in \{1,\dots,N\}$.

Then the dynamics defined in (10) *satisfies*

$$\frac{1}{T}\int_0^T \mathbb{E}[\mathrm{KGD}_k^2(Q_N^t)]\,\mathrm{d}t \le \frac{C_N}{T} + \frac{C_k}{N}, \quad C_N := \mathrm{KLD}(\mu_0||Q_0) + \mathbb{E}_{\theta_1,\dots,\theta_N\sim\mu_0}[\mathcal{L}(Q_N)] - \mathcal{L}_{\min},$$

for some constant C_k *which is* N *and* T*-independent.*

Proof. The proof is organised into four steps:

Step 1: Flow of the particle system. Introduce the shorthand $\boldsymbol{\theta} := (\theta_1,\dots,\theta_N) \in \mathbb{R}^{d\times N}$ and

$$\Phi_{\boldsymbol{\theta}}(\theta_i,\theta_j) := k(\theta_i,\theta_j)\underbrace{(\nabla\log q_0 - \nabla_{\mathrm{V}}\mathcal{L}(Q_N))(\theta_j)}_{=:b_{Q_N}(\theta_j)} + \nabla_1 k(\theta_j,\theta_i), \qquad Q_N := \frac{1}{N}\sum_{j=1}^N \delta_{\theta_j},$$

where for convenience we have suppressed the t-dependence, i.e. $\theta_i \equiv \theta_i(t)$ and $Q_N \equiv Q_N(\boldsymbol{\theta})$. Let $\boldsymbol{V} = (V_1,\dots,V_N) : \mathbb{R}^{d\times N} \to \mathbb{R}^{d\times N}$ with $V_i(\boldsymbol{\theta}) := \frac{1}{N}\sum_{j=1}^N \Phi_{\boldsymbol{\theta}}(\theta_i,\theta_j)$, so that (10) reads $\dot{\boldsymbol{\theta}} = \boldsymbol{V}(\boldsymbol{\theta})$. Under our assumptions $\boldsymbol{V}$ is $C^1(\mathbb{R}^{d\times N})$: indeed, k and $\nabla_1 k$ are C^1 by (v), $\nabla\log q_0$ is C^1 by (iii), and $\boldsymbol{\theta} \mapsto \nabla_{\mathrm{V}}\mathcal{L}(Q_N)(\theta_j)$ is $C^1(\mathbb{R}^{d\times N})$ by (ii), composing $(\boldsymbol{\theta},\vartheta) \mapsto \nabla_{\mathrm{V}}\mathcal{L}(Q_N)(\vartheta)$ with $\boldsymbol{\theta} \mapsto (\boldsymbol{\theta},\theta_j)$. Further, from (vi), $\Phi_{\boldsymbol{\theta}}(\theta_i,\theta_j)$ has at most linear growth as a function of $\boldsymbol{\theta}$; i.e. $\|\Phi_{\boldsymbol{\theta}}(\theta_i,\theta_j)\| \lesssim 1 + \|\boldsymbol{\theta}\|$, and hence $\|\boldsymbol{V}(\boldsymbol{\theta})\| \lesssim 1 + \|\boldsymbol{\theta}\|$. Since $\boldsymbol{V}$ is C^1 with at most

linear growth, solutions of $\dot{\boldsymbol{\theta}} = \boldsymbol{V}(\boldsymbol{\theta})$ exist, are unique, and are defined for all $t \in \mathbb{R}$, and the associated flow $\boldsymbol{F}_t : \mathbb{R}^{d\times N} \to \mathbb{R}^{d\times N}$, $\boldsymbol{F}_0 = \mathrm{Id}$, is a C^1 diffeomorphism for each t (with inverse $\boldsymbol{F}_{-t}$). Further, $(t,\boldsymbol{\eta}) \mapsto \boldsymbol{F}_t(\boldsymbol{\eta})$ and $(t,\boldsymbol{\eta}) \mapsto D\boldsymbol{F}_t(\boldsymbol{\eta})$ are continuous, and the Jacobian matrix satisfies the variational equation $\partial_t D\boldsymbol{F}_t(\boldsymbol{\eta}) = D\boldsymbol{V}(\boldsymbol{F}_t(\boldsymbol{\eta}))\, D\boldsymbol{F}_t(\boldsymbol{\eta})$ [Hartman, 2002, Chapter 5, Theorem 3.1]. Let $\Lambda_t(\boldsymbol{\eta}) := \det D\boldsymbol{F}_t(\boldsymbol{\eta})$; from the variational equation and Jacobi's formula,

$$\partial_t \log \Lambda_t(\boldsymbol{\eta}) = \mathrm{tr}\big(D\boldsymbol{V}(\boldsymbol{F}_t(\boldsymbol{\eta}))\big) = (\nabla\cdot\boldsymbol{V})(\boldsymbol{F}_t(\boldsymbol{\eta})), \qquad \Lambda_0(\boldsymbol{\eta}) = 1, \tag{18}$$

so that $\Lambda_t(\boldsymbol{\eta}) = \exp(\int_0^t (\nabla\cdot\boldsymbol{V})(\boldsymbol{F}_s(\boldsymbol{\eta}))\,\mathrm{d}s) > 0$. Since $\boldsymbol{\theta}(t) = \boldsymbol{F}_t(\boldsymbol{\theta}(0))$ and $\boldsymbol{\theta}(0)$ has density $p_N(0,\cdot) = \mu_0^{\otimes N}$, the change of variables formula shows that $\boldsymbol{\theta}(t)$ has a density $p_N(t,\cdot)$, characterised by

$$p_N(t, \boldsymbol{F}_t(\boldsymbol{\eta}))\, \Lambda_t(\boldsymbol{\eta}) = p_N(0,\boldsymbol{\eta}), \qquad \boldsymbol{\eta} \in \mathbb{R}^{d\times N}. \tag{19}$$

Let $S_0 := \operatorname{supp} p_N(0,\cdot) = (\operatorname{supp}\mu_0)^N$, which is compact by (iv). Then $p_N(t,\cdot)$ is supported in $\boldsymbol{F}_t(S_0)$ and, for $t \in [0,T]$, these sets are contained in the compact set $\{\boldsymbol{F}_s(\boldsymbol{\eta}) : s \in [0,T],\, \boldsymbol{\eta} \in S_0\}$. Note that no differentiability of p_N will be required.

Step 2: Descent on the N-particle free energy.

Consider the function $\pi_N : \mathbb{R}^{d\times N} \to (0,\infty)$ defined as

$$\pi_N(\boldsymbol{\theta}) := q_0(\theta_1)\cdots q_0(\theta_N) \exp\left(-N\mathcal{L}(Q_N)\right), \qquad Q_N \equiv Q_N(\boldsymbol{\theta}), \tag{20}$$

which is (up to normalisation) the N-particle Gibbs measure associated with the objective $\mathcal{J}(Q) = \mathcal{L}(Q) + \mathrm{KLD}(Q||Q_0)$ [cf. Chen et al., 2022]. Crucially, π_N is a fixed function on $\mathbb{R}^{d\times N}$; it does not depend on t. From (i), $\boldsymbol{\theta} \mapsto \log\pi_N(\boldsymbol{\theta})$ is $C^1(\mathbb{R}^{d\times N})$ and its gradient can be computed exactly, with all dependence of Q_N on $\boldsymbol{\theta}$ accounted for, as

$$\nabla_{\theta_i} \log\pi_N(\boldsymbol{\theta}) = \nabla \log q_0(\theta_i) - N\nabla_{\theta_i}\mathcal{L}(Q_N) = \nabla\log q_0(\theta_i) - \nabla_{\mathrm{V}}\mathcal{L}(Q_N)(\theta_i) = b_{Q_N}(\theta_i). \tag{21}$$

Let

$$H(t) := \int \log\left(\frac{p_N(t,\boldsymbol{\theta})}{\pi_N(\boldsymbol{\theta})}\right) p_N(t,\boldsymbol{\theta})\,\mathrm{d}\boldsymbol{\theta} = \mathrm{KLD}(p_N(t,\cdot)||Q_0^{\otimes N}) + N\,\mathbb{E}\left[\mathcal{L}(Q_N)\right], \tag{22}$$

where the expectation is with respect to $\boldsymbol{\theta} \sim p_N(t,\cdot)$. Since π_N does not depend on t, the t dependence of $H(t)$ enters only through the term $p_N(t,\boldsymbol{\theta})$ in the integrand.

Write $p_0 \equiv p_N(0,\cdot)$. Here $H(0)$ is finite; see (26) below. The change of variables $\boldsymbol{\theta} = \boldsymbol{F}_t(\boldsymbol{\eta})$, together with (19), gives

$$\begin{aligned} H(t) &= \int \log\left(\frac{p_0(\boldsymbol{\eta})}{\pi_N(\boldsymbol{F}_t(\boldsymbol{\eta}))\,\Lambda_t(\boldsymbol{\eta})}\right) p_0(\boldsymbol{\eta})\,\mathrm{d}\boldsymbol{\eta} \\ &= H(0) - \int \left[\log\pi_N(\boldsymbol{F}_t(\boldsymbol{\eta})) - \log\pi_N(\boldsymbol{\eta}) + \log\Lambda_t(\boldsymbol{\eta})\right] p_0(\boldsymbol{\eta})\,\mathrm{d}\boldsymbol{\eta}, \end{aligned} \tag{23}$$

with the convention $0 \log 0 = 0$. From the chain rule, $\partial_t \log \pi_N(\boldsymbol{F}_t(\boldsymbol{\eta})) = \nabla \log \pi_N(\boldsymbol{F}_t(\boldsymbol{\eta})) \cdot \boldsymbol{V}(\boldsymbol{F}_t(\boldsymbol{\eta}))$, which together with (18) shows that the t-derivative of the integrand in (23) is $p_0(\boldsymbol{\eta})$ multiplied by a function that is continuous in $(t, \boldsymbol{\eta})$, and hence bounded on the compact set $[0, T] \times S_0$. Differentiation under the integral is therefore justified by the dominated convergence theorem and, transforming back using (19),

$$\begin{aligned} H'(t) &= -\int \left[(\nabla \cdot \boldsymbol{V})(\boldsymbol{F}_t(\boldsymbol{\eta})) + \nabla \log \pi_N(\boldsymbol{F}_t(\boldsymbol{\eta})) \cdot \boldsymbol{V}(\boldsymbol{F}_t(\boldsymbol{\eta})) \right] p_0(\boldsymbol{\eta}) \, \mathrm{d}\boldsymbol{\eta} \\ &= -\int \left[(\nabla \cdot \boldsymbol{V})(\boldsymbol{\theta}) + \nabla \log \pi_N(\boldsymbol{\theta}) \cdot \boldsymbol{V}(\boldsymbol{\theta}) \right] p_N(t, \boldsymbol{\theta}) \, \mathrm{d}\boldsymbol{\theta}. \end{aligned}$$

Finally, $(\nabla \cdot \boldsymbol{V})(\boldsymbol{\theta}) = \sum_{i=1}^N \nabla_{\theta_i} \cdot V_i(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N \nabla_{\theta_i} \cdot \Phi_{\boldsymbol{\theta}}(\theta_i, \theta_j)$, where ∇_{θ_i} acts on all dependence on θ_i, and from (21) we have $\nabla \log \pi_N(\boldsymbol{\theta}) \cdot \boldsymbol{V}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N b_{Q_N}(\theta_i) \cdot \Phi_{\boldsymbol{\theta}}(\theta_i, \theta_j)$. Thus

$$\begin{aligned} H'(t) &= -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N \int (\nabla_{\theta_i} \cdot \Phi_{\boldsymbol{\theta}}(\theta_i, \theta_j) + b_{Q_N}(\theta_i) \cdot \Phi_{\boldsymbol{\theta}}(\theta_i, \theta_j)) p_N(t, \boldsymbol{\theta}) \, \mathrm{d}\boldsymbol{\theta} \\ &= -\mathbb{E}\left[\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N \nabla_{\theta_i} \cdot \Phi_{\boldsymbol{\theta}}(\theta_i, \theta_j) + b_{Q_N}(\theta_i) \cdot \Phi_{\boldsymbol{\theta}}(\theta_i, \theta_j) \right] \end{aligned} \tag{24}$$

where we have used the expectation shorthand to refer to the random initialisation of the particles.

Step 3: Calculating derivatives. Now we aim to calculate the terms in (24):

$$\begin{aligned} \nabla_{\theta_i} \cdot \Phi_{\boldsymbol{\theta}}(\theta_i, \theta_j) &= \nabla_{\theta_i} \cdot [k(\theta_i, \theta_j) b_{Q_N}(\theta_j)] + \nabla_{\theta_i} \cdot [\nabla_1 k(\theta_j, \theta_i)] \\ &= \nabla_1 k(\theta_i, \theta_j) \cdot b_{Q_N}(\theta_j) + k(\theta_i, \theta_j) \nabla_{\theta_i} \cdot b_{Q_N}(\theta_j) + \nabla_1 \cdot \nabla_2 k(\theta_i, \theta_j) \\ &\quad + \{\nabla_1 k(\theta_i, \theta_i) \cdot b_{Q_N}(\theta_i) + \Delta_1 k(\theta_i, \theta_i)\} \mathbb{1}_{i=j} \\ b_{Q_N}(\theta_i) \cdot \Phi_{\boldsymbol{\theta}}(\theta_i, \theta_j) &= k(\theta_i, \theta_j) b_{Q_N}(\theta_i) \cdot b_{Q_N}(\theta_j) + b_{Q_N}(\theta_i) \cdot \nabla_1 k(\theta_j, \theta_i) \end{aligned}$$

and, writing $\Psi(\boldsymbol{\theta}, \vartheta) := \nabla_{\mathrm{V}} \mathcal{L}(Q_N)(\vartheta)$, the multivariate chain rule gives $\nabla_{\theta_i} \cdot [\Psi(\boldsymbol{\theta}, \theta_j)] = \{(\nabla_\vartheta \cdot \Psi)(\boldsymbol{\theta}, \theta_i)\} \mathbb{1}_{i=j} + [\nabla_{\theta_i} \cdot \Psi(\boldsymbol{\theta}, \vartheta)]_{\vartheta=\theta_j}$, where the first term is $\nabla \cdot \nabla_{\mathrm{V}} \mathcal{L}(Q_N)(\theta_i)$ and, from the second chain rule in (ii), the second term is $\frac{1}{N} \nabla_{\mathrm{V}} \cdot \nabla_{\mathrm{V}} \mathcal{L}(Q_N)(\theta_j)(\theta_i)$, so that

$$\nabla_{\theta_i} \cdot b_{Q_N}(\theta_j) = \{\nabla \cdot (\nabla \log q_0 - \nabla_{\mathrm{V}} \mathcal{L}(Q_N))(\theta_i)\} \mathbb{1}_{i=j} - \frac{1}{N} \nabla_{\mathrm{V}} \cdot \nabla_{\mathrm{V}} \mathcal{L}(Q_N)(\theta_j)(\theta_i).$$

Thus, collecting together terms that correspond to KGD using (17),

$$\begin{aligned} H'(t) = -N\mathbb{E}\left[\mathrm{KGD}_k^2(Q_N)\right] + \mathbb{E}\left[\frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N k(\theta_i, \theta_j) \nabla_{\mathrm{V}} \cdot \nabla_{\mathrm{V}} \mathcal{L}(Q_N)(\theta_j)(\theta_i) \right] \\ - \mathbb{E}\left[\frac{1}{N} \sum_{i=1}^N k(\theta_i, \theta_i) [\Delta \log q_0(\theta_i) - \nabla \cdot \nabla_{\mathrm{V}} \mathcal{L}(Q_N)(\theta_i)] + \Delta_1 k(\theta_i, \theta_i) + \nabla_1 k(\theta_i, \theta_i) \cdot b_{Q_N}(\theta_i) \right]. \end{aligned} \tag{25}$$

Step 4: Obtaining a bound. The final task is to bound the non-KGD terms appearing in (25) by a Q_N-independent constant. Under our assumptions,

$$\begin{aligned}
&\left|\frac{1}{N^2}\sum_{i=1}^{N}\sum_{j=1}^{N} k(\theta_i,\theta_j)\nabla_{\mathrm{V}}\cdot\nabla_{\mathrm{V}}\mathcal{L}(Q_N)(\theta_j)(\theta_i)\right| \\
&\qquad \leq \sup_{Q\in\mathcal{P}(\mathbb{R}^d)}\left|\iint k(\theta,\vartheta)\nabla_{\mathrm{V}}\cdot\nabla_{\mathrm{V}}\mathcal{L}(Q)(\theta)(\vartheta)\,\mathrm{d}Q(\theta)\mathrm{d}Q(\vartheta)\right| < \infty \\
&\left|\frac{1}{N}\sum_{i=1}^{N} k(\theta_i,\theta_i)[\Delta\log q_0(\theta_i) - \nabla\cdot\nabla_{\mathrm{V}}\mathcal{L}(Q_N)(\theta_i)] + \Delta_1 k(\theta_i,\theta_i) + \nabla_1 k(\theta_i,\theta_i)\cdot b_{Q_N}(\theta_i)\right| \\
&\qquad \leq \sup_{\theta} k(\theta,\theta)|\Delta\log q_0(\theta)| \\
&\qquad\quad + \sup_{Q\in\mathcal{P}(\mathbb{R}^d)}\left|\int k(\theta,\theta)\nabla\cdot\nabla_{\mathrm{V}}\mathcal{L}(Q)(\theta)\,\mathrm{d}Q(\theta)\right| + \sup_{\theta}|\Delta_1 k(\theta,\theta)| \\
&\qquad\quad + \sup_{\theta}|\nabla_1 k(\theta,\theta)\cdot\nabla\log q_0(\theta)| \\
&\qquad\quad + \sup_{Q\in\mathcal{P}(\mathbb{R}^d)}\left|\int \nabla_1 k(\theta,\theta)\cdot\nabla_{\mathrm{V}}\mathcal{L}(Q)(\theta)\,\mathrm{d}Q(\theta)\right| < \infty
\end{aligned}$$

which establish the required bounds, i.e.

$$H'(t) \leq -N\mathbb{E}[\mathrm{KGD}_k^2(Q_N)] + C_k$$

for some finite, k-dependent constant C_k. Integrating both sides from 0 to T and rearranging yields

$$\frac{1}{T}\int_0^T \mathbb{E}[\mathrm{KGD}_k^2(Q_N)]\,\mathrm{d}t \leq \frac{H(0)-H(T)}{NT} + \frac{C_k}{N}.$$

It remains to bound $H(0)-H(T)$. Since $Q_0^{\otimes N}$ is a probability distribution, the first term in (22) is non-negative, while from (i) the second term is bounded below by $N\mathcal{L}_{\min}$, so that

$$H(T) \geq N\mathcal{L}_{\min}.$$

At $t=0$ we have $p_N(0,\cdot) = \mu_0^{\otimes N}$ and, using the tensorisation of the KLD over product measures,

$$\begin{aligned}
H(0) &= \mathrm{KLD}(\mu_0^{\otimes N}||Q_0^{\otimes N}) + N\,\mathbb{E}_{\theta_1,\ldots,\theta_N\sim\mu_0}[\mathcal{L}(Q_N)] \\
&= N\left\{\mathrm{KLD}(\mu_0||Q_0) + \mathbb{E}_{\theta_1,\ldots,\theta_N\sim\mu_0}[\mathcal{L}(Q_N)]\right\}. \qquad (26)
\end{aligned}$$

Here $\mathrm{KLD}(\mu_0||Q_0) < \infty$, since from (iv) the density μ_0 is bounded, so that $\int \mu_0\log\mu_0 \leq \log\sup_\theta \mu_0(\theta)$, while from (iii) q_0 is continuous and positive, and hence bounded away from zero on the compact support of μ_0. Combining these bounds, $H(0)-H(T) \leq NC_N$, as required.

□

Remark 9 (Relaxing the conditions of Proposition 3 in Chazal et al. [2025])**.** *Compared to Proposition 3 in Chazal et al. [2025] we have relaxed both (ii) on the loss and condition (v) on the kernel. In (ii) we have relaxed boundedness conditions on $\nabla\nabla_{\mathrm{V}}\mathcal{L}$ and $\nabla_{\mathrm{V}}\cdot\nabla_{\mathrm{V}}\mathcal{L}$ (which do not hold for $\mathcal{L}_{\mathrm{PrO}}$ in our context) into integrability conditions on $\nabla\cdot\nabla_{\mathrm{V}}\mathcal{L}$ and $\nabla_{\mathrm{V}}\cdot\nabla_{\mathrm{V}}\mathcal{L}$ (which, as we will see, do hold under appropriate assumptions on P_θ). In condition (v) we have also relaxed the assumption that the kernel is translation-invariant.*

A.3.2 Verifying Regularity Conditions

Proposition 5 applied to general loss functions $\mathcal{L}$; here we establish explicit sufficient conditions in the specific case of $\mathcal{L}_{\mathrm{PrO}}$.

Proposition 6 (Regularity for the variational gradient of $\mathcal{L}_{\mathrm{PrO}}$)**.** *Let $p_\theta(y_i|x_i)$ be a positive density for all θ and each (x_i, y_i) in the dataset. Let*

(i) $\sup_\theta p_\theta(y_i|x_i) < \infty$

(ii) $\sup_\theta \sqrt{k(\theta,\theta)}\dfrac{\|\nabla_\theta p_\theta(y_i|x_i)\|}{p_\theta(y_i|x_i)} < \infty$

(iii) $\sup_\theta k(\theta,\theta)\dfrac{|\Delta_\theta p_\theta(y_i|x_i)|}{p_\theta(y_i|x_i)} < \infty$

(iv) $\sup_\theta \|\nabla_1 k(\theta,\theta)\|\dfrac{\|\nabla_\theta p_\theta(y_i|x_i)\|}{p_\theta(y_i|x_i)} < \infty$

for each (x_i, y_i) in the dataset. Then the map $\theta \mapsto k(\theta,\theta)\|\nabla_\theta p_\theta(y_i|x_i)\|$ has at most linear growth and, for the loss function $\mathcal{L} = \mathcal{L}_{\mathrm{PrO}}$ in (5)*,*

$$\sup_{Q\in\mathcal{P}(\mathbb{R}^d)}\left|\int k(\theta,\theta)\nabla\cdot\nabla_{\mathrm{V}}\mathcal{L}(Q)(\theta)\,\mathrm{d}Q(\theta)\right| < \infty$$

$$\sup_{Q\in\mathcal{P}(\mathbb{R}^d)}\left|\iint k(\theta,\vartheta)\nabla_{\mathrm{V}}\cdot\nabla_{\mathrm{V}}\mathcal{L}(Q)(\theta)(\vartheta)\,\mathrm{d}Q(\theta)\mathrm{d}Q(\vartheta)\right| < \infty$$

$$\sup_{Q\in\mathcal{P}(\mathbb{R}^d)}\left|\int \nabla_1 k(\theta,\theta)\cdot\nabla_{\mathrm{V}}\mathcal{L}(Q)(\theta)\,\mathrm{d}Q(\theta)\right| < \infty.$$

Proof. The first claim follows from combining (i) and (ii). Continuing from Proposition 2, the second variation is

$$\mathcal{L}''(Q)(\theta)(\vartheta) = \sum_{i=1}^n \frac{p_\theta(y_i|x_i)p_\vartheta(y_i|x_i)}{p_Q(y_i|x_i)^2},$$

which is well-defined since, under our assumptions, $p_Q(y_i|x_i) = \int p_\theta(y_i|x_i)\,\mathrm{d}Q(\theta)$ is strictly positive for all $Q \in \mathcal{P}(\mathbb{R}^d)$ and all (x_i, y_i) in the dataset. The required variational gradients

are

$$\nabla \cdot \nabla_{\mathrm{V}} \mathcal{L}(Q)(\theta) = -\sum_{i=1}^{n} \frac{\Delta_\theta p_\theta(y_i|x_i)}{p_Q(y_i|x_i)}$$
$$\nabla_{\mathrm{V}} \cdot \nabla_{\mathrm{V}} \mathcal{L}(Q)(\theta)(\vartheta) = \sum_{i=1}^{n} \frac{\nabla_\theta p_\theta(y_i|x_i) \cdot \nabla_\vartheta p_\vartheta(y_i|x_i)}{p_Q(y_i|x_i)^2},$$

whose terms we assumed to be well-defined. Finally, since each $k(\theta,\theta)\Delta_\theta p_\theta(y_i|x_i)$ is bounded in θ, $k(\theta,\theta)\nabla \cdot \nabla_{\mathrm{V}} \mathcal{L}(Q)$ is Q-integrable for each $Q \in \mathcal{P}(\mathbb{R}^d)$. Likewise, since $|k(\theta,\vartheta)| \leq \sqrt{k(\theta,\theta)}\sqrt{k(\vartheta,\vartheta)}$ and each $\sqrt{k(\theta,\theta)}\nabla_\theta p_\theta(y_i|x_i)$ is bounded in θ, we deduce that $k(\theta,\vartheta)\nabla_{\mathrm{V}} \cdot \nabla_{\mathrm{V}} \mathcal{L}(Q)(\theta)(\vartheta)$ is $Q \otimes Q$-integrable for each $Q \in \mathcal{P}(\mathbb{R}^d)$. Integrating these equations and applying Jensen's inequality,

$$\left| \int k(\theta,\theta) \nabla \cdot \nabla_{\mathrm{V}} \mathcal{L}(Q)(\theta) \, \mathrm{d}Q(\theta) \right| \leq \sum_{i=1}^{n} \frac{\int k(\theta,\theta) |\Delta_\theta p_\theta(y_i|x_i)| \, \mathrm{d}Q(\theta)}{\int p_\theta(y_i|x_i) \, \mathrm{d}Q(\theta)}$$
$$\left| \iint k(\theta,\vartheta) \nabla_{\mathrm{V}} \cdot \nabla_{\mathrm{V}} \mathcal{L}(Q)(\theta)(\vartheta) \, \mathrm{d}Q(\theta) \mathrm{d}Q(\vartheta) \right| \leq \sum_{i=1}^{n} \left(\frac{\int \sqrt{k(\theta,\theta)} \|\nabla_\theta p_\theta(y_i|x_i)\| \, \mathrm{d}Q(\theta)}{\int p_\theta(y_i|x_i) \, \mathrm{d}Q(\theta)} \right)^2.$$

Let Π_Q denote the distribution for which $(\mathrm{d}\Pi_Q/\mathrm{d}Q)(\theta) \propto p_\theta(y_i|x_i)$, so that

$$\frac{\int k(\theta,\theta) |\Delta_\theta p_\theta(y_i|x_i)| \, \mathrm{d}Q(\theta)}{\int p_\theta(y_i|x_i) \, \mathrm{d}Q(\theta)} = \int \underbrace{k(\theta,\theta) \frac{|\Delta_\theta p_\theta(y_i|x_i)|}{p_\theta(y_i|x_i)}}_{(*)} \, \mathrm{d}\Pi_Q(\theta)$$
$$\frac{\int \sqrt{k(\theta,\theta)} \|\nabla_\theta p_\theta(y_i|x_i)\| \, \mathrm{d}Q(\theta)}{\int p_\theta(y_i|x_i) \, \mathrm{d}Q(\theta)} = \int \underbrace{\sqrt{k(\theta,\theta)} \frac{\|\nabla_\theta p_\theta(y_i|x_i)\|}{p_\theta(y_i|x_i)}}_{(**)} \, \mathrm{d}\Pi_Q(\theta)$$

where, under our assumptions, both integrands $(*)$ and $(**)$ are bounded over $\theta \in \mathbb{R}^d$. It follows that both integrals are bounded over $\Pi_Q \in \mathcal{P}(\mathbb{R}^d)$, and hence over $Q \in \mathcal{P}(\mathbb{R}^d)$. Similarly, from (15),

$$\left| \int \nabla_1 k(\theta,\theta) \cdot \nabla_{\mathrm{V}} \mathcal{L}(Q)(\theta) \, \mathrm{d}Q(\theta) \right| \leq \sum_{i=1}^{n} \int \|\nabla_1 k(\theta,\theta)\| \frac{\|\nabla_\theta p_\theta(y_i|x_i)\|}{p_\theta(y_i|x_i)} \, \mathrm{d}\Pi_Q(\theta),$$

where the integrand is bounded over $\theta \in \mathbb{R}^d$ by (iv), so that this integral is also bounded over $Q \in \mathcal{P}(\mathbb{R}^d)$, completing the argument. □

A.3.3 Proof of Theorem 2

At last we can present a proof of Theorem 2:

Proof of Theorem 2. Our first task is to verify the conditions of Proposition 5 for $\mathcal{L} = \mathcal{L}_{\mathrm{PrO}}$:

(i) (Lower bound and chain rule) Since $\theta \mapsto p_\theta(y_i|x_i)$ is bounded for each (x_i, y_i), $p_Q(y_i|x_i) \leq \sup_\theta p_\theta(y_i|x_i)$ for all $Q \in \mathcal{P}(\mathbb{R}^d)$, and thus

$$\mathcal{L}_{\min} \geq -\sum_{i=1}^n \log \sup_\theta p_\theta(y_i|x_i) > -\infty.$$

Further, $\mathcal{L}(Q_N) = -\sum_{i=1}^n \log(\frac{1}{N}\sum_{r=1}^N p_{\theta_r}(y_i|x_i))$ is $C^2(\mathbb{R}^{d\times N})$, since each $\theta \mapsto p_\theta(y_i|x_i)$ is positive and $C^2(\mathbb{R}^d)$, and direct differentiation together with (15) gives

$$\nabla_{\theta_j}\mathcal{L}(Q_N) = -\sum_{i=1}^n \frac{\frac{1}{N}\nabla_{\theta_j} p_{\theta_j}(y_i|x_i)}{\frac{1}{N}\sum_{r=1}^N p_{\theta_r}(y_i|x_i)} = \frac{1}{N}\nabla_{\mathrm{V}}\mathcal{L}(Q_N)(\theta_j).$$

Finally, by convexity of $z \mapsto -\log(z)$,

$$\begin{aligned}\mathcal{L}(Q_N) &\leq \frac{1}{N}\sum_{r=1}^N \left[-\sum_{i=1}^n \log p_{\theta_r}(y_i|x_i)\right] \\ \Rightarrow \quad \mathbb{E}_{\theta_1,\ldots,\theta_N\sim\mu_0}[\mathcal{L}(Q_N)] &\leq -\sum_{i=1}^n \int \log p_\theta(y_i|x_i)\, \mathrm{d}\mu_0(\theta), \end{aligned} \tag{27}$$

which is finite since $\theta \mapsto p_\theta(y_i|x_i)$ is continuous and positive, and hence bounded away from zero on the bounded support of μ_0.

(ii) (Loss) Since each $\theta \mapsto p_\theta(y_i|x_i)$ is positive and $C^2(\mathbb{R}^d)$, from Proposition 2 the first variation $\mathcal{L}'(Q)$ is $C^2(\mathbb{R}^d)$ for each $Q \in \mathcal{P}(\mathbb{R}^d)$, and $\nabla_{\mathrm{V}} \cdot \nabla_{\mathrm{V}}\mathcal{L}(Q)$ is well-defined. Moreover, $(\theta_1, \ldots, \theta_N) \mapsto p_{Q_N}(y_i|x_i)$ is positive and $C^2(\mathbb{R}^{d\times N})$, so from (15) the map $(\boldsymbol{\theta}, \vartheta) \mapsto \nabla_{\mathrm{V}}\mathcal{L}(Q_N)(\vartheta) = -\sum_{i=1}^n \nabla_\vartheta p_\vartheta(y_i|x_i)/p_{Q_N}(y_i|x_i)$ is $C^1(\mathbb{R}^{d\times N}\times\mathbb{R}^d)$, and direct differentiation gives

$$\begin{aligned}\nabla_{\theta_r} \cdot [\nabla_{\mathrm{V}}\mathcal{L}(Q_N)(\vartheta)] &= \frac{1}{N}\sum_{i=1}^n \frac{\nabla_\vartheta p_\vartheta(y_i|x_i)\cdot\nabla_{\theta_r} p_{\theta_r}(y_i|x_i)}{p_{Q_N}(y_i|x_i)^2} \\ &= \frac{1}{N}\nabla_{\mathrm{V}} \cdot \nabla_{\mathrm{V}}\mathcal{L}(Q_N)(\vartheta)(\theta_r),\end{aligned}$$

which is the second chain rule required in (ii). Further, from (15) and the fact that $\frac{1}{N}\sum_{r=1}^N p_{\theta_r}(y_i|x_i) \geq \frac{1}{N}p_{\theta_j}(y_i|x_i)$,

$$\begin{aligned}|\nabla_{\mathrm{V}}\mathcal{L}(Q_N)(\theta_j)| &= \left|-\sum_{i=1}^n \frac{\nabla_{\theta_j} p_{\theta_j}(y_i|x_i)}{\frac{1}{N}\sum_{r=1}^N p_{\theta_r}(y_i|x_i)}\right| \\ &\leq N\sum_{i=1}^n \left|\frac{\nabla_{\theta_j} p_{\theta_j}(y_i|x_i)}{p_{\theta_j}(y_i|x_i)}\right| = N\sum_{i=1}^n \|\nabla_{\theta_j}\log p_{\theta_j}(y_i|x_i)\|.\end{aligned}$$

From Proposition 6 all three of the integrability conditions on the loss in (ii) of Proposition 5 are satisfied.

(iii) (Regularisation) Satisfied by assumption.

(iv) (Initialisation) Satisfied by assumption.

(v) (Kernel) Satisfied by assumption.

(vi) (Growth) Let $s := \max_i \sup_\theta \sqrt{k(\theta,\theta)}\|\nabla_\theta \log p_\theta(y_i|x_i)\|$, which is finite by condition (iv)a. of Theorem 2. From the bound in (ii) and the bound $|k(\theta_i,\theta_j)| \leq \sqrt{k(\theta_i,\theta_i)}\sqrt{k(\theta_j,\theta_j)}$,

$$\begin{aligned}|k(\theta_i,\theta_j)|\|\nabla_{\mathrm{V}}\mathcal{L}(Q_N)(\theta_j)\| &\leq N\sum_{\ell=1}^{n}\sqrt{k(\theta_i,\theta_i)}\left[\sqrt{k(\theta_j,\theta_j)}\|\nabla_{\theta_j}\log p_{\theta_j}(y_\ell|x_\ell)\|\right] \\ &\leq Nns\sqrt{k(\theta_i,\theta_i)}.\end{aligned}$$

Since k is symmetric, $\nabla_\theta[k(\theta,\theta)] = 2\nabla_1 k(\theta,\theta)$, which has at most linear growth by condition (ii) of Theorem 2; thus $k(\theta,\theta) \lesssim 1 + \|\theta\|^2$, and $\sqrt{k(\theta_i,\theta_i)}$, and hence $|k(\theta_i,\theta_j)|\|\nabla_{\mathrm{V}}\mathcal{L}(Q_N)(\theta_j)\|$, has at most linear growth. The remaining growth requirements were directly assumed.

It remains only to establish that the constant C_N appearing in the conclusion of Proposition 5 can be bounded by an N-independent constant C_0. Combining the lower bound on $\mathcal{L}_{\min}$ with (27),

$$\begin{aligned}C_N &= \mathrm{KLD}(\mu_0||Q_0) + \mathbb{E}_{\theta_1,\ldots,\theta_N\sim\mu_0}[\mathcal{L}(Q_N)] - \mathcal{L}_{\min} \\ &\leq \mathrm{KLD}(\mu_0||Q_0) + \sum_{i=1}^{n}\int \log\left(\frac{\sup_\vartheta p_\vartheta(y_i|x_i)}{p_\theta(y_i|x_i)}\right)\mathrm{d}\mu_0(\theta) =: C_0,\end{aligned}$$

which is finite, non-negative, and independent of both N and T. This completes the argument. □

A.4 Proof of Proposition 1

This appendix is dedicated to a proof of our final theoretical result:

Proof of Proposition 1. For these calculation we recall that $A : B = \mathrm{tr}(AB^\top)$ for matrices A, B is the double dot product and that $[\nabla_\theta v(\theta)]_{i,j} = \nabla_{\theta_i} v_j(\theta)$ and $[\Delta_\theta v(\theta)]_j = \Delta_\theta v_j(\theta)$ for vector-valued $v : \mathbb{R}^d \to \mathbb{R}^d$. By convention $v(\theta)$ is a column vector and $\Delta_\theta v(\theta)$ is a row vector for $v : \mathbb{R}^d \to \mathbb{R}^d$. For the Gaussian location model

$$\begin{aligned}\nabla_\theta p_\theta(y_i|x_i) &= (\nabla_\theta f_\theta(x_i))\Sigma^{-1}(y_i - f_\theta(x_i))p_\theta(y_i|x_i) \\ \Delta_\theta p_\theta(y_i|x_i) &= (\Delta_\theta f_\theta(x_i))\Sigma^{-1}(y_i - f_\theta(x_i))p_\theta(y_i|x_i) \\ &\qquad + (\nabla_\theta f_\theta(x_i)) : [\nabla_\theta p_\theta(y_i|x_i)(y_i - f_\theta(x_i))^\top - (\nabla_\theta f_\theta(x_i))p_\theta(y_i|x_i)]\Sigma^{-1} \\ &= \left\{\begin{array}{l}(\Delta_\theta f_\theta(x_i))\Sigma^{-1}(y_i - f_\theta(x_i)) \\ \quad +(\nabla_\theta f_\theta(x_i)) : [(\nabla_\theta f_\theta(x_i))\Sigma^{-1}(y_i - f_\theta(x_i))(y_i - f_\theta(x_i))^\top\Sigma^{-1}] \\ \quad -(\nabla_\theta f_\theta(x_i)) : [(\nabla_\theta f_\theta(x_i))\Sigma^{-1}]\end{array}\right\} p_\theta(y_i|x_i)\end{aligned}$$

so that

$$\sup_\theta \sqrt{k(\theta,\theta)}\frac{\|\nabla_\theta p_\theta(y_i|x_i)\|}{p_\theta(y_i|x_i)} = \sup_\theta \sqrt{k(\theta,\theta)}\|(\nabla_\theta f_\theta(x_i))\Sigma^{-1}(y_i - f_\theta(x_i))\|$$
$$\leq \|\Sigma^{-1}y_i\| \sup_\theta \sqrt{k(\theta,\theta)}\|(\nabla_\theta f_\theta(x_i))\|_{\mathrm{op}}$$
$$+ \|\Sigma^{-1}\|_{\mathrm{op}} \left[\sup_\theta \|f_\theta(x_i)\|\right] \left[\sup_\theta \sqrt{k(\theta,\theta)}\|(\nabla_\theta f_\theta(x_i))\|_{\mathrm{op}}\right]$$

where the finiteness of the terms appearing on the right hand side was assumed. An identical argument, with $\sqrt{k(\theta,\theta)}$ replaced by $\|\nabla_1 k(\theta,\theta)\|$, shows that condition (iv)c. of Theorem 2 is satisfied. A similar but more lengthy calculation (which we omit for brevity) for the supremum in condition (iv)b. completes the argument. □

A.5 Proof of Theorem 3 and Corollary 1

This appendix is devoted to the proofs of Theorem 3 and Corollary 1. First we present three auxiliary results; the equivariance of the posteriors and the test statistic for linear models (Proposition 7), the stability of entropy-regularised minimisers with respect to perturbations of the reference distribution (Proposition 8), and control of the MMD between mixtures using total variation (Proposition 9). Then we present the proofs of Theorem 3 and Corollary 1.

For $Q_0 \in \mathcal{P}(\mathbb{R}^d)$ and a dataset $\mathfrak{D} = \{(x_i, y_i)\}_{i=1}^n$, let $Q_{\mathrm{Bayes}}^{\mathfrak{D},Q_0}$ and $Q_{\mathrm{PrO}}^{\mathfrak{D},Q_0}$ denote the minimisers of (3) with $\mathcal{L} = \mathcal{L}_{\mathrm{Bayes}}^{\mathfrak{D}}$ and $\mathcal{L} = \mathcal{L}_{\mathrm{PrO}}^{\mathfrak{D}}$ respectively, where the superscripts make the dependence on the dataset and the prior explicit, and let $P_{\mathrm{Bayes}}^{\mathfrak{D},Q_0}$ and $P_{\mathrm{PrO}}^{\mathfrak{D},Q_0}$ denote the associated predictive distributions. Under assumptions (i) and (ii) of Theorem 3, these minimisers exist and are unique, by Theorem 1 for Q_{PrO} and by the variational characterisation of the Bayesian posterior [Zellner, 1988] for Q_{Bayes}.

For $c \in \mathbb{R}^p$ let $s_c(y) := y + c$, and for $\Delta \in \mathbb{R}^d$ let $\tau_\Delta(\theta) := \theta - \Delta$.

Proposition 7 (Equivariance for Linear Models)**.** *Assume that condition (i) of Theorem 3 holds. Let $\mathfrak{D} = \{(x_i, y_i)\}_{i=1}^n$, let $\Delta \in \mathbb{R}^d$, and let $\mathfrak{D}' = \{(x_i, y_i + A(x_i)\Delta)\}_{i=1}^n$. Then, for* $\mathrm{M} \in \{\mathrm{Bayes}, \mathrm{PrO}\}$*,*

$$Q_{\mathrm{M}}^{\mathfrak{D}',Q_0} = (\tau_\Delta^{-1})_\# Q_{\mathrm{M}}^{\mathfrak{D},(\tau_\Delta)_\# Q_0} \qquad \textit{and} \qquad P_{\mathrm{M}}^{\mathfrak{D}',Q_0}(\cdot|x) = (s_{A(x)\Delta})_\# P_{\mathrm{M}}^{\mathfrak{D},(\tau_\Delta)_\# Q_0}(\cdot|x)$$

for all x.

Proof. From condition (i), $p_\vartheta(y + A(x)\Delta|x) = g_x(y - A(x)(\vartheta - \Delta)) = p_{\tau_\Delta(\vartheta)}(y|x)$ for all ϑ, x and y. Thus, for any $Q \in \mathcal{P}(\mathbb{R}^d)$,

$$\mathcal{L}_{\mathrm{PrO}}^{\mathfrak{D}'}(Q) = -\sum_{i=1}^n \log \int p_{\tau_\Delta(\vartheta)}(y_i|x_i)\,\mathrm{d}Q(\vartheta)$$
$$= -\sum_{i=1}^n \log \int p_\theta(y_i|x_i)\,\mathrm{d}((\tau_\Delta)_\# Q)(\theta) = \mathcal{L}_{\mathrm{PrO}}^{\mathfrak{D}}((\tau_\Delta)_\# Q),$$

and an identical argument shows that $\mathcal{L}^{\mathfrak{D}'}_{\mathrm{Bayes}}(Q) = \mathcal{L}^{\mathfrak{D}}_{\mathrm{Bayes}}((\tau_\Delta)_\# Q)$. Since the KLD is invariant under the application of a common bijection to both of its arguments, $\mathrm{KLD}(Q||Q_0) = \mathrm{KLD}((\tau_\Delta)_\# Q||(\tau_\Delta)_\# Q_0)$. It follows that the objective (3) based on $\mathfrak{D}'$ and Q_0, evaluated at Q, is equal to the objective based on $\mathfrak{D}$ and $(\tau_\Delta)_\# Q_0$, evaluated at $(\tau_\Delta)_\# Q$. Since $Q \mapsto (\tau_\Delta)_\# Q$ is a bijection on $\mathcal{P}(\mathbb{R}^d)$ and the minimisers are unique, the first claim is established. For the second claim, note from condition (i) that $P_\vartheta(\cdot|x) = (s_{A(x)\Delta})_\# P_{\tau_\Delta(\vartheta)}(\cdot|x)$, and thus, using the first claim,

$$P^{\mathfrak{D}',Q_0}_{\mathrm{M}}(\cdot|x) = \int P_\vartheta(\cdot|x)\,\mathrm{d}Q^{\mathfrak{D}',Q_0}_{\mathrm{M}}(\vartheta) = (s_{A(x)\Delta})_\# \int P_\theta(\cdot|x)\,\mathrm{d}Q^{\mathfrak{D},(\tau_\Delta)_\# Q_0}_{\mathrm{M}}(\theta),$$

as claimed. □

Proposition 8 (Stability with Respect to the Prior)**.** *Let $\mathcal{L} : \mathcal{P}(\mathbb{R}^d) \to (-\infty, \infty]$ be convex and bounded below. Let $\pi_1, \pi_2 \in \mathcal{P}(\mathbb{R}^d)$ be mutually absolutely continuous with $h := \log(\mathrm{d}\pi_2/\mathrm{d}\pi_1)$ satisfying $\sup_\theta |h(\theta)| \leq \delta$. For $k \in \{1, 2\}$ suppose that $\mathcal{J}_k(Q) := \mathcal{L}(Q) + \mathrm{KLD}(Q||\pi_k)$ has a minimiser R_k with $\mathcal{J}_k(R_k) < \infty$. Then $\|R_1 - R_2\|_{\mathrm{TV}} \leq 2\delta$.*

Proof. First we show that $\mathcal{J}_k(Q) - \mathcal{J}_k(R_k) \geq \mathrm{KLD}(Q||R_k)$ for all Q with $\mathcal{J}_k(Q) < \infty$, i.e. that $\mathcal{J}_k$ is 1-strongly convex with respect to the KLD at its minimiser. Fix such a Q, and for $t \in (0, 1)$ let $R_t := (1-t)R_k + tQ$. Since $\mathcal{L}$ is bounded below, $\mathrm{KLD}(R_k||\pi_k)$ and $\mathrm{KLD}(Q||\pi_k)$ are finite, and direct calculation with densities with respect to π_k gives the identity

$$(1-t)\mathrm{KLD}(R_k||\pi_k) + t\mathrm{KLD}(Q||\pi_k) = \mathrm{KLD}(R_t||\pi_k) + (1-t)\mathrm{KLD}(R_k||R_t) + t\mathrm{KLD}(Q||R_t).$$

Combining this identity with the convexity of $\mathcal{L}$ and the optimality of R_k,

$$\mathcal{J}_k(R_k) \leq \mathcal{J}_k(R_t) \leq (1-t)\mathcal{J}_k(R_k) + t\mathcal{J}_k(Q) - (1-t)\mathrm{KLD}(R_k||R_t) - t\mathrm{KLD}(Q||R_t),$$

and rearranging gives $\mathcal{J}_k(Q) - \mathcal{J}_k(R_k) \geq \mathrm{KLD}(Q||R_t)$. Since $R_t \to R_k$ in total variation as $t \to 0$, the lower semicontinuity of the KLD gives $\mathcal{J}_k(Q) - \mathcal{J}_k(R_k) \geq \mathrm{KLD}(Q||R_k)$, as claimed.

Now, since $\mathrm{KLD}(Q||\pi_1) = \mathrm{KLD}(Q||\pi_2) + \int h\,\mathrm{d}Q$ whenever either side is finite, and h is bounded, we have $\mathcal{J}_1(Q) = \mathcal{J}_2(Q) + \int h\,\mathrm{d}Q$, and in particular $\mathcal{J}_1(R_2) < \infty$ and $\mathcal{J}_2(R_1) < \infty$. Applying the first part of the proof twice,

$$\begin{aligned}\mathrm{KLD}(R_2||R_1) + \mathrm{KLD}(R_1||R_2) &\leq \{\mathcal{J}_1(R_2) - \mathcal{J}_1(R_1)\} + \{\mathcal{J}_2(R_1) - \mathcal{J}_2(R_2)\} \\ &= \int h\,\mathrm{d}(R_2 - R_1) \leq \delta\|R_1 - R_2\|_{\mathrm{TV}} \leq \delta\sqrt{2\mathrm{KLD}(R_2||R_1)},\end{aligned}$$

where the last step used Pinsker's inequality. Thus $\mathrm{KLD}(R_2||R_1) \leq 2\delta^2$, and a final application of Pinsker's inequality gives $\|R_1 - R_2\|_{\mathrm{TV}} \leq \sqrt{2\mathrm{KLD}(R_2||R_1)} \leq 2\delta$. □

Proposition 9 (Controlling MMD by Total Variation)**.** *Assume that condition (iii) of Theorem 3 holds. Then, for all x and all $R, R' \in \mathcal{P}(\mathbb{R}^d)$,*

$$\mathrm{MMD}_\kappa\left(\int P_\theta(\cdot|x)\,\mathrm{d}R(\theta), \int P_\theta(\cdot|x)\,\mathrm{d}R'(\theta)\right) \leq M\|R - R'\|_{\mathrm{TV}}.$$

Moreover, $\mathrm{MMD}_\kappa(s_{c\#}P, s_{c\#}P') = \mathrm{MMD}_\kappa(P, P')$ for all $c \in \mathbb{R}^p$ and all distributions P, P'.

Proof. For any distribution P, $\|\mu_P\|^2_{\mathcal{H}_\kappa} = \iint \kappa(y, y') \,\mathrm{d}P(y)\mathrm{d}P(y') \leq M^2$, since $|\kappa(y, y')| \leq \sqrt{\kappa(y, y)\kappa(y', y')} \leq M^2$. The kernel mean embedding of the mixture $\int P_\theta(\cdot|x) \,\mathrm{d}R(\theta)$ is the Bochner integral $\int \mu_{P_\theta(\cdot|x)} \,\mathrm{d}R(\theta)$, and thus

$$\begin{aligned}\mathrm{MMD}_\kappa\left(\int P_\theta(\cdot|x) \,\mathrm{d}R(\theta), \int P_\theta(\cdot|x) \,\mathrm{d}R'(\theta)\right) &= \left\|\int \mu_{P_\theta(\cdot|x)} \,\mathrm{d}(R - R')(\theta)\right\|_{\mathcal{H}_\kappa} \\ &\leq M \,\|R - R'\|_{\mathrm{TV}},\end{aligned}$$

which establishes the first claim. For the second claim, translation-invariance of κ gives

$$\mathrm{MMD}^2_\kappa(s_{c\#}P, s_{c\#}P') = \iint \kappa(y + c, y' + c) \,\mathrm{d}(P - P')(y)\mathrm{d}(P - P')(y') = \mathrm{MMD}^2_\kappa(P, P'),$$

as claimed. □

A.5.1 Proof of Theorem 3

Proof of Theorem 3. Fix θ, ϑ, u, and the covariates, and let $\Delta := \vartheta - \theta$. Let $\mathfrak{D}$ and $\mathfrak{D}'$ denote the datasets generated by (13) using, respectively, (θ, u) and (ϑ, u). From condition (i), $G(\vartheta, x_i, u) = G(\theta, x_i, u) + A(x_i)\Delta$, so that $\mathfrak{D}$ and $\mathfrak{D}'$ are related as in Proposition 7. Let $\tilde{Q}_0 := (\tau_\Delta)_\# Q_0$. From Proposition 7, the predictive distributions based on $\mathfrak{D}'$ and Q_0 are obtained by translating the predictive distributions based on $\mathfrak{D}$ and $\tilde{Q}_0$ by $A(x_i)\Delta$, and this translation is common to the Bayesian and PrO predictive distributions. Thus, from the second part of Proposition 9,

$$\begin{aligned}\mathcal{T}_n(\vartheta, u) &= \frac{1}{n}\sum_{i=1}^n \mathrm{MMD}^2_\kappa\left(P^{\mathfrak{D},\tilde{Q}_0}_{\mathrm{PrO}}(\cdot|x_i), P^{\mathfrak{D},\tilde{Q}_0}_{\mathrm{Bayes}}(\cdot|x_i)\right) \\ \mathcal{T}_n(\theta, u) &= \frac{1}{n}\sum_{i=1}^n \mathrm{MMD}^2_\kappa\left(P^{\mathfrak{D},Q_0}_{\mathrm{PrO}}(\cdot|x_i), P^{\mathfrak{D},Q_0}_{\mathrm{Bayes}}(\cdot|x_i)\right);\end{aligned}$$

that is, the two test statistics are computed from the same dataset, and differ only in the location of the prior. For a prior π, let

$$d_i(\pi) := \mu_{P^{\mathfrak{D},\pi}_{\mathrm{PrO}}(\cdot|x_i)} - \mu_{P^{\mathfrak{D},\pi}_{\mathrm{Bayes}}(\cdot|x_i)} \in \mathcal{H}_\kappa.$$

Then $\sqrt{\mathcal{T}_n(\theta, u)}$ is the norm of $(d_1(Q_0), \ldots, d_n(Q_0))$ in the Hilbert space $\mathcal{H}^n_\kappa$ equipped with the inner product $\langle (a_i), (b_i)\rangle := \frac{1}{n}\sum_{i=1}^n \langle a_i, b_i\rangle_{\mathcal{H}_\kappa}$, and similarly $\sqrt{\mathcal{T}_n(\vartheta, u)}$ is the norm of $(d_1(\tilde{Q}_0), \ldots, d_n(\tilde{Q}_0))$. From the reverse triangle inequality in this space, the triangle inequality in $\mathcal{H}_\kappa$, and the first part of Proposition 9,

$$\begin{aligned}\left|\sqrt{\mathcal{T}_n(\vartheta, u)} - \sqrt{\mathcal{T}_n(\theta, u)}\right| &\leq \max_{i=1,\ldots,n} \|d_i(\tilde{Q}_0) - d_i(Q_0)\|_{\mathcal{H}_\kappa} \\ &\leq M\left\{\|Q^{\mathfrak{D},\tilde{Q}_0}_{\mathrm{PrO}} - Q^{\mathfrak{D},Q_0}_{\mathrm{PrO}}\|_{\mathrm{TV}} + \|Q^{\mathfrak{D},\tilde{Q}_0}_{\mathrm{Bayes}} - Q^{\mathfrak{D},Q_0}_{\mathrm{Bayes}}\|_{\mathrm{TV}}\right\}.\end{aligned}$$

Finally we apply Proposition 8 with $\pi_1 = Q_0$ and $\pi_2 = \tilde{Q}_0$, for which $h(\theta) = \log q_0(\theta + \Delta) - \log q_0(\theta)$ satisfies $\sup_\theta |h(\theta)| \leq L_0\|\Delta\|$ from condition (ii). The losses $\mathcal{L}_{\mathrm{PrO}}$ and $\mathcal{L}_{\mathrm{Bayes}}$ are both convex, and bounded below by $-\sum_{i=1}^n \log \sup_y g_{x_i}(y) > -\infty$ from condition (i). Moreover, the minimal values of the objectives are finite for any prior π with a positive density; for $\mathcal{L}_{\mathrm{PrO}}$ the objective evaluated at π itself is $-\sum_{i=1}^n \log \int p_\theta(y_i|x_i)\,\mathrm{d}\pi(\theta)$, which is finite since the integrands are positive and bounded, while for $\mathcal{L}_{\mathrm{Bayes}}$ the minimal value is $-\log \int \prod_{i=1}^n p_\theta(y_i|x_i)\,\mathrm{d}\pi(\theta)$, which is finite for the same reason. Thus Proposition 8 applies to both posteriors, giving

$$\left|\sqrt{\mathcal{T}_n(\vartheta, u)} - \sqrt{\mathcal{T}_n(\theta, u)}\right| \leq M\left\{2L_0\|\Delta\| + 2L_0\|\Delta\|\right\} = 4ML_0\|\vartheta - \theta\|,$$

which establishes the first claim.

For the second claim, fix θ and ϑ, and let $\epsilon := K\|\vartheta - \theta\|$. Let F_θ denote the distribution function of $\sqrt{\mathcal{T}_n(\theta, U)}$, $U \sim \nu$, and let $q_\theta := \inf\{t : F_\theta(t) \geq 1-\alpha\}$ denote its $(1-\alpha)$-quantile; similarly for ϑ. Since $\sqrt{\mathcal{T}_n(\vartheta, u)} \leq \sqrt{\mathcal{T}_n(\theta, u)} + \epsilon$ for all u, we have $F_\vartheta(t + \epsilon) \geq F_\theta(t)$ for all t, and setting $t = q_\theta$ and using right-continuity of F_θ gives $F_\vartheta(q_\theta + \epsilon) \geq 1 - \alpha$, so that $q_\vartheta \leq q_\theta + \epsilon$. Exchanging the roles of θ and ϑ gives $|q_\vartheta - q_\theta| \leq \epsilon$, as claimed. □

A.5.2 Proof of Corollary 1

Proof of Corollary 1. Let $q_n(\theta)$ denote the $(1-\alpha)$-quantile of $\sqrt{\mathcal{T}_n(\theta, U)}$, $U \sim \nu$, so that the bootstrap critical value (in the idealised setting where infinitely many bootstrap replicates are used) is $\hat{c}_n = q_n(\theta_n)^2$, and the test rejects if and only if $\sqrt{r_n \mathcal{T}_n(\theta_\star, u)} > \sqrt{r_n}\, q_n(\theta_n)$. From Theorem 3,

$$|\sqrt{r_n}\, q_n(\theta_n) - \sqrt{r_n}\, q_n(\theta_\star)| \leq K\sqrt{r_n}\|\theta_n - \theta_\star\| \xrightarrow{p} 0.$$

Let $Z \sim F$ and let c_α denote the $(1-\alpha)$-quantile of F. From the continuous mapping theorem, $\sqrt{r_n \mathcal{T}_n(\theta_\star, U)} \xrightarrow{d} \sqrt{Z}$, and the distribution function $s \mapsto F(s^2)$ of $\sqrt{Z}$ is continuous and strictly increasing at its $(1-\alpha)$-quantile $\sqrt{c_\alpha}$. Since $\sqrt{r_n}\, q_n(\theta_\star)$ is the $(1-\alpha)$-quantile of $\sqrt{r_n \mathcal{T}_n(\theta_\star, U)}$, convergence of quantiles [Lemma 21.2 of van der Vaart, 1998] gives $\sqrt{r_n}\, q_n(\theta_\star) \to \sqrt{c_\alpha}$, and hence $\sqrt{r_n}\, q_n(\theta_n) \xrightarrow{p} \sqrt{c_\alpha}$. Under the null, the observed test statistic is $\mathcal{T}_n(\theta_\star, u)$ with $u \sim \nu$, and θ_n is computed from the same data. Although these two quantities are dependent, $\sqrt{r_n}\, q_n(\theta_n)$ converges in probability to a constant, so that from Slutsky's theorem

$$\sqrt{r_n \mathcal{T}_n(\theta_\star, u)} - \sqrt{r_n}\, q_n(\theta_n) \xrightarrow{d} \sqrt{Z} - \sqrt{c_\alpha}.$$

Since F is continuous at c_α, $\mathbb{P}(\sqrt{Z} = \sqrt{c_\alpha}) = 0$, and the portmanteau theorem gives

$$\mathbb{P}\left(\mathcal{T}_n(\theta_\star, u) > \hat{c}_n\right) \to \mathbb{P}(Z > c_\alpha) = 1 - F(c_\alpha) = \alpha,$$

as claimed. □

B Experimental Protocol

This appendix contains the details required to reproduce the experiments reported in Section 3. The test problems that we consider are specified in Section B.1, details of the maximum mean discrepancy test statistic are contained in Section B.2, implementational aspects of VGD are discussed in Section B.3, and additional empirical results are contained in Section B.4. Full details for the seismic travel time tomography case study are contained in Section B.6.

Code Code to reproduce our simulation study from Section 3.1 is available at `https://github.com/liuqingyang27/Detecting-Model-Misspecification-via-VGD`. Code to reproduce our seismic travel time tomography experiments from Section 3.2 is available at `https://github.com/XuebinZhaoZXB/Detecting-Model-Misspecification/`.

B.1 Test Problems

The regression functions that we considered for our simulation study in Section 3.1 are as follows:

1. $f_\theta(x) = \theta x^2$ with $\theta \in \mathbb{R}$ and $\{x_i\}$ uniformly sampled from $[0, 1]$
2. $f_\theta(x) = \frac{1}{1+e^{-\theta x}}$ with $\theta \in \mathbb{R}$ and $\{x_i\}$ uniformly sampled from $[-1, 1]$
3. $f_\theta(x) = \theta_1 + \theta_2 x$ with $\theta \in \mathbb{R}^2$ and $\{x_i\}$ uniformly sampled from $[-2, 2]$

In the well-specified scenario, data are generated by $y_i = f_\theta(x_i) + z_i$ for all $i \in \{1, \cdots, n\}$, where z_i are i.i.d $\mathcal{N}(0, \sigma^2)$. For the three cases above, the noise levels are separately $0.5, 0.05$ and 0.8. The true data-parameters parameters in this case were $\theta = 5$ for the quadratic model, $\theta = 5$ for the sigmoid model and $\theta = [5, 3]$ for the linear model. To generate data that are misspecified we proceeded as follows for each of the above tasks:

1. $y_i = (5 + 3u_i)x_i^2 + z_i$ for all $i \in \{1, \cdots, n\}$, where $u_i \sim \mathcal{N}(0, 1),\ z_i \sim \mathcal{N}(0, 0.5^2)$
2. data points are generated from a uniform distribution with density

$$f(x, y) = \begin{cases} \frac{1}{2}, & \text{if } (x, y) \in (0, 1) \times (0, 1), \\ \frac{1}{2}, & \text{if } (x, y) \in (-1, 0) \times (-1, 0), \\ 0, & \text{otherwise.} \end{cases}$$

3. $y_i = 5 + 3x_i + 2x_i^2 + z_i$ for all $i \in \{1, \cdots, n\}$, where $z_i \sim \mathcal{N}(0, 0.8^2)$.

In the second of the above examples $\theta \mapsto f_\theta(x)$, $\nabla_\theta f_\theta(x)$ and $\Delta_\theta f_\theta(x)$ are bounded, so from Proposition 1 the sufficient conditions of our theory are satisfied whenever the kernel k is bounded. On the other hand, in the first and third cases our theoretical assumptions are *not* satisfied; this enables us to test the performance of Algorithm 1 outside the setting where our theoretical results hold.

B.2 Maximum Mean Discrepancy

The maximum mean discrepancy employed in these experiments utilised a Gaussian kernel

$$\kappa(y, y') = \exp\left(-\frac{\|y - y'\|^2}{2\ell^2}\right)$$

where the lengthscale was selected as the standard deviation of $\{y_i\}_{i=1}^n$. For the synthetic examples we present in Section 3.1, the dimension of the response variable is always $p = 1$, but for completeness here we work with the general form of the Gaussian kernel. The choice of a Gaussian kernel together with the Gaussian measurement error model (14) with covariance matrix $\Sigma = \sigma^2 I_{p\times p}$ enables (12) to be explicitly computed using the analytic form of the integral

$$\begin{aligned}\mathfrak{k}(\theta, \vartheta|x_i) &:= \iint \exp\left(-\frac{\|y - y'\|^2}{2\ell^2}\right) \mathrm{d}P_\theta(y|x_i)\mathrm{d}P_\vartheta(y'|x_i) \\ &= \left(\frac{\ell^2}{\ell^2 + 2\sigma^2}\right)^{p/2} \exp\left(-\frac{\|f_\theta(x_i) - f_\vartheta(x_i)\|^2}{2(\ell^2 + 2\sigma^2)}\right)\end{aligned}$$

together with

$$\begin{aligned}\mathrm{MMD}^2(P_{\mathrm{Bayes}}(\cdot|x_i), P_{\mathrm{PrO}}(\cdot|x_i)) = &\iint \mathfrak{k}(\theta, \vartheta|x_i)\ \mathrm{d}Q_{\mathrm{Bayes}}(\theta)\mathrm{d}Q_{\mathrm{Bayes}}(\vartheta) \\ &- 2\iint \mathfrak{k}(\theta, \vartheta|x_i)\ \mathrm{d}Q_{\mathrm{Bayes}}(\theta)\mathrm{d}Q_{\mathrm{PrO}}(\vartheta) \\ &+ \iint \mathfrak{k}(\theta, \vartheta|x_i)\ \mathrm{d}Q_{\mathrm{PrO}}(\theta)\mathrm{d}Q_{\mathrm{PrO}}(\vartheta). \end{aligned} \tag{28}$$

In practice both Q_{Bayes} and Q_{PrO} are approximated using VGD, so we obtain a particle-based representation $\{\theta_i^{\mathrm{Bayes}}\}_{i=1}^N$ for Q_{Bayes} and $\{\theta_i^{\mathrm{PrO}}\}_{i=1}^N$ for Q_{PrO}. Substituting these empirical measures into (28) we obtain

$$\begin{aligned}\mathrm{MMD}^2(P_{\mathrm{Bayes}}(\cdot|x_i), P_{\mathrm{PrO}}(\cdot|x_i)) \approx &\frac{1}{N^2}\sum_{r=1}^N\sum_{s=1}^N \mathfrak{k}(\theta_r^{\mathrm{Bayes}}, \theta_s^{\mathrm{Bayes}}|x_i) \\ &- 2\frac{1}{N^2}\sum_{r=1}^N\sum_{s=1}^N \mathfrak{k}(\theta_r^{\mathrm{Bayes}}, \theta_s^{\mathrm{PrO}}|x_i) \\ &+ \frac{1}{N^2}\sum_{r=1}^N\sum_{s=1}^N \mathfrak{k}(\theta_r^{\mathrm{PrO}}, \theta_s^{\mathrm{PrO}}|x_i). \end{aligned} \tag{29}$$

The approximate values in (29) were used for the experiments that we report in Section 3 of the main text.

B.3 Variational Gradient Descent

For our toy experiments we utilised the inverse multiquadric kernel

$$k(\theta, \vartheta) = \left(c^2 + \frac{\|\theta - \vartheta\|^2}{l^2}\right)^{-\beta}$$

with $\beta = 0.5$. To select an appropriate length scale l, we employed the median heuristic [Garreau et al., 2017] at each iteration of Algorithm 1. The step size and iteration number in each experiment were manually selected to ensure convergence, as quantified by KGD (see e.g. Figure 5). In all experiments $N = 20$ particles were used.

B.4 Additional Empirical Results

The posterior distributions Q_{Bayes} and Q_{PrO} corresponding to the regression tasks in Figure 2 are displayed in Figure 5, alongside the values of the KGD in (11) obtained along the trajectory of VGD. A kernel density estimator has been applied to the particle representations of Q_{Bayes} and Q_{PrO} to aid visualisation in Figure 5. It can be seen that the standard Bayesian posterior Q_{Bayes} is rather concentrated in all scenarios, irrespective of whether the statistical model is well-specified or misspecified, while Q_{PrO} tends to be more diffuse when the statistical model is misspecified. The values of KGD obtained along the trajectory of VGD appear to generally decrease and converge to a limit in all cases, consistent with an accurate N-particle approximation having been found.

Further to the analysis in the main text, we investigate the performance of our method under different data sizes n and different dimensions of θ.

First, in Figure 6, we considered the sigmoid regression task with varying sample sizes of $n = 100, 1000$ and 10000. The test statistic values $\mathcal{T}(\{(x_i, \tilde{y}_i)\}_{i=1}^n)$ calculated under the (bootstrap) null typically decrease as the data size grows, while under misspecification the actual $\mathcal{T}$ values are effectively n-independent. Consequently, misspecified models are easier to detect when we have a larger dataset, as would be expected.

Second, in Figure 7 we consider a regression model defined by

$$f_\theta(x) = \sum_{p=1}^{d} \theta_p \sin(px). \tag{30}$$

For the misspecified scenario, the data are generated according to $y_i = \sin(1/x_i) + z_i$ where $z_i \sim \mathcal{N}(0, \sigma^2)$ and $\sigma = 0.2$, a function that remains misspecified regardless of the dimension p of the model (30). In particular, we cannot expect (30) to resolve the rapid oscillations around $x = 0$. For presentational purposes we consider $p \in \{5, 20, 50\}$. The predictive distributions P_{Bayes} and P_{PrO} perform generally well when the model is well-specified. In the misspecified case, P_{Bayes} is over-confident around $x = 0$ and this is partially remedied in P_{PrO}. The power of diagnostic declines with increasing parameter dimension p, as the actual MMD value gets closer to the effective support of the null; however, the test still had power to reject the well-specified null even when $p = 50$.

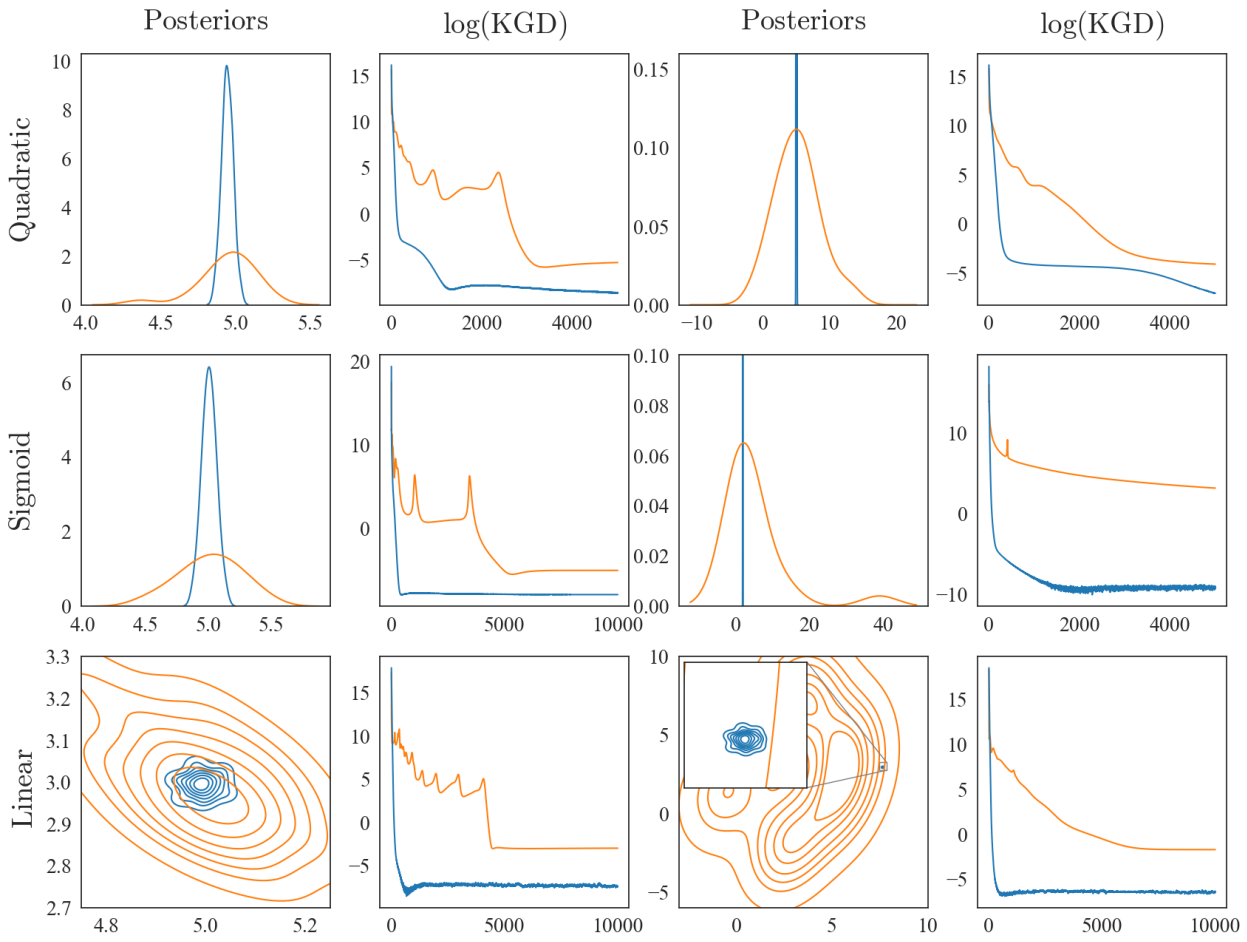


Figure 5: Simulation Study. Each row corresponds to a regression task in Figure 2 in which the data are either generated from the statistical model (well-specified, left) or not generated from the statistical model (misspecified, right). The posterior distributions Q_{Bayes} (blue) and Q_{PrO} (orange) are displayed, together with the values of the KGD in (11) obtained along the trajectory of VGD.

In summary, our additional experiments confirm the intuition that larger dataset sizes n increase our power to detect when the model is misspecified, while larger parameter dimension p decreases our power to detect when the model is misspecified.

B.5 Insensitivity to the Learning Rate

This appendix demonstrates the sensitivity analysis described in Remark 6. For each of the three regression tasks in Section B.1, in both the well-specified and the misspecified settings, we computed the parametric bootstrap p-value p_λ for each $\lambda \in \Lambda = \{2^{-3}, 2^{-2}, 2^{-1}, 1, 2, 2^2, 2^3\}$, using 100 bootstrap replicates. The results are displayed in Figure 8. In the well-specified setting p_λ is not small for any λ, while in the misspecified setting the null is rejected for all

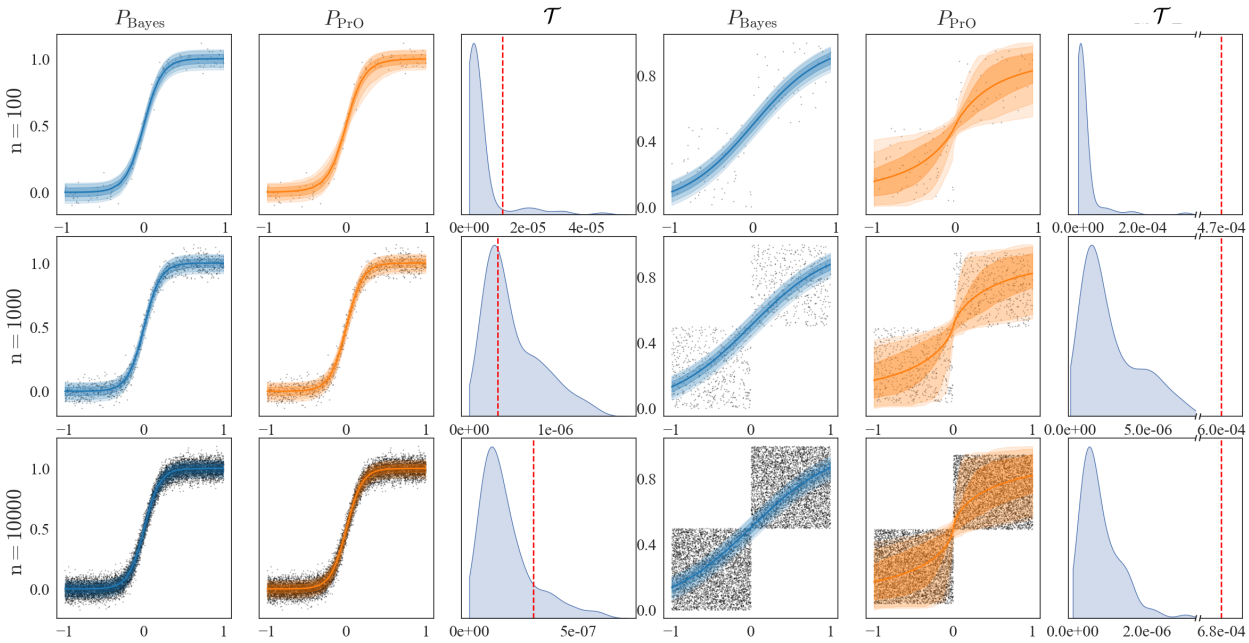


Figure 6: Additional simulation study, varying the size n of the dataset. Each row corresponds to the sigmoid regression task in Figure 2 with different data sizes in which the data are either generated from the statistical model (well-specified, left) or not generated from the statistical model (misspecified, right). The posterior predictive distributions P_{Bayes} and P_{PrO} are displayed, along with the null distribution under the hypothesis that the statistical model is well-specified, and actual realised value of the test statistic $\mathcal{T}$ in (12) (red dashed).

$\lambda \in \Lambda$. In the quadratic and linear scenarios, p-values are not sensitive to the learning rate.

B.6 Details for Seismic Travel Time Tomography

For computation using VGD, a set of $N = 600$ particles $\{\theta_j^0\}_{j=1}^N$ were initialised by sampling from the prior Q_0. A total of $T = 500$ iterations of VGD were performed with step size $\epsilon = 0.1$. The Gaussian kernel was used, in line with earlier work in this context, and the length scale was calculated by the median of pairwise distances between particles [Garreau et al., 2017].

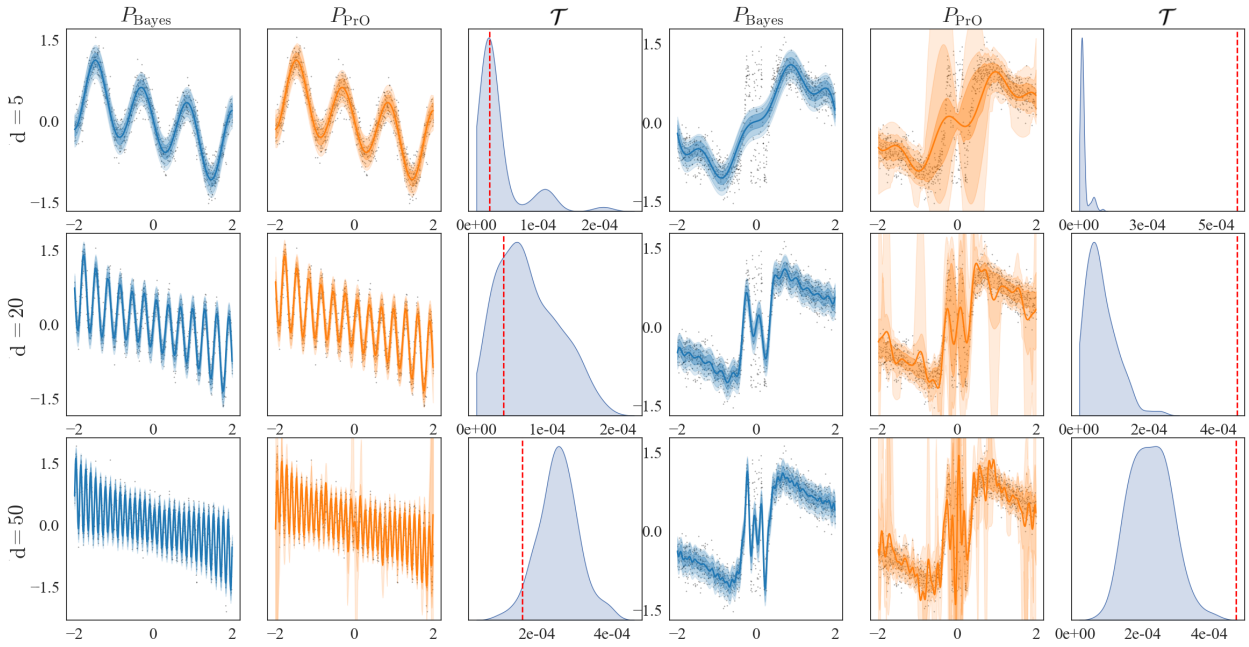


Figure 7: Additional simulation study, varying the number d of parameters in the model. Each row corresponds to a regression task using model (30) with different parameters dimension, in which the data are either generated from the statistical model (well-specified, left) or not generated from the statistical model (misspecified, right). The posterior predictive distributions P_{Bayes} and P_{PrO} are displayed, along with the null distribution under the hypothesis that the statistical model is well-specified, and actual realised value of the test statistic $\mathcal{T}$ in (12) (red dashed).

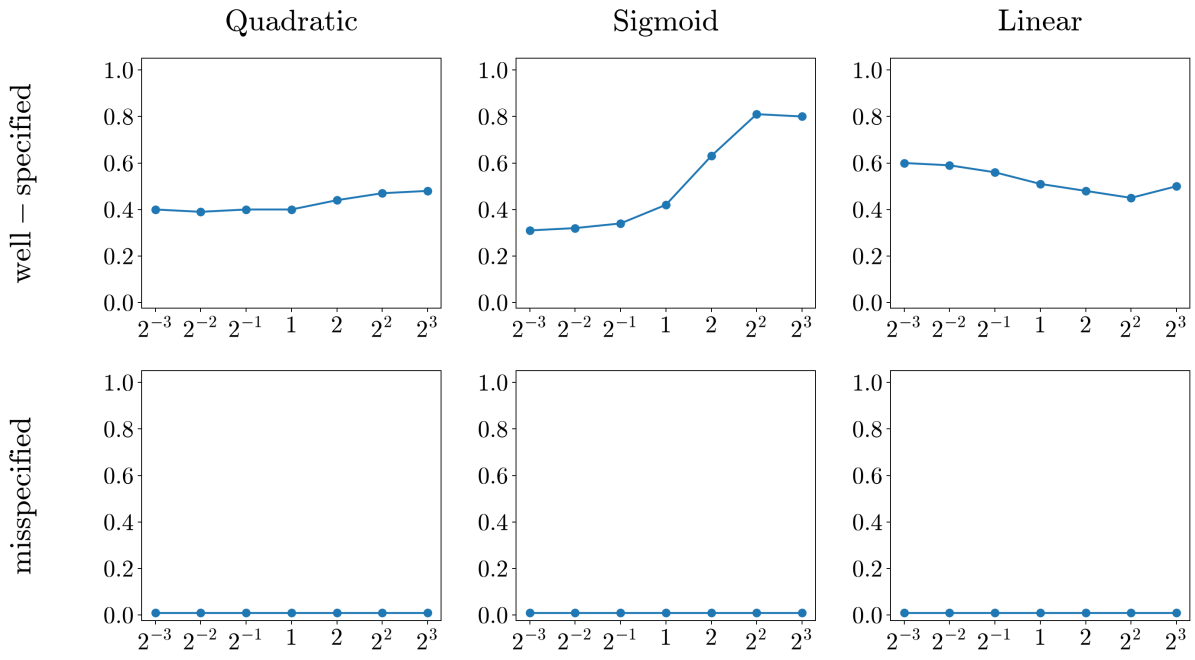


Figure 8: Insensitivity to the learning rate. The parametric bootstrap p-value p_λ is displayed as a function of the learning rate λ, for each of the regression tasks in Figure 2, in the well-specified (top) and misspecified (bottom) settings.